



(12) **United States Patent**
McNaughton

(10) **Patent No.:** **US 12,409,915 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **WAKE SHAPING APPARATUS AND
RELATED TECHNOLOGY**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 513 days.

(21) Appl. No.: **17/718,238**

(22) Filed: **Apr. 11, 2022**

(65) **Prior Publication Data**

US 2022/0234696 A1 Jul. 28, 2022

Related U.S. Application Data

(63) Continuation-in-part of application No. 16/251,742,
filed on Jan. 18, 2019, now Pat. No. 11,299,241,
which is a continuation of application No.
15/858,779, filed on Dec. 29, 2017, now Pat. No.
10,183,726.

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(51) **Int. Cl.**
B63B 34/75 (2020.01)
B63B 1/00 (2006.01)
B63B 1/32 (2006.01)
B63B 34/70 (2020.01)

(52) **U.S. Cl.**
CPC **B63B 34/75** (2020.02); **B63B 1/32**
(2013.01); **B63B 2001/005** (2013.01); **B63B**
2001/325 (2013.01); **B63B 34/70** (2020.02);
Y02T 70/10 (2013.01)

(58) **Field of Classification Search**

CPC ... B63B 34/70; B63B 34/75; B63B 2001/005;
B63B 2001/281; B63B 1/30; B63B 1/32;
B63B 2001/325; B63B 39/06; B63B
39/061; B63B 2039/065

See application file for complete search history.

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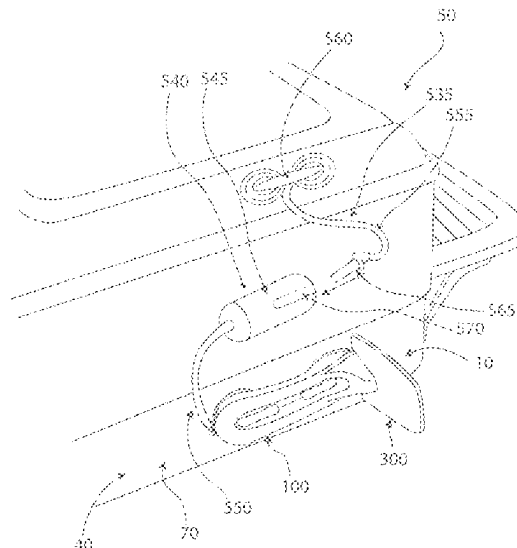
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P.A.

(57) **ABSTRACT**

The invention provides an apparatus that attaches to a hull
of a boat to disrupt water flow around the hull such that a
wake created behind the boat is larger, differently posi-
tioned, and/or differently shaped. In some embodiments, the
apparatus is removably attached to the hull by one or more
suction cups. The wake shaping apparatus can include a
blade and be configured such that an angular orientation of
the blade is adjustable. In certain embodiments, the wake
shaping apparatus is inflatable.

17 Claims, 43 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/551,621, filed on Aug. 29, 2017.

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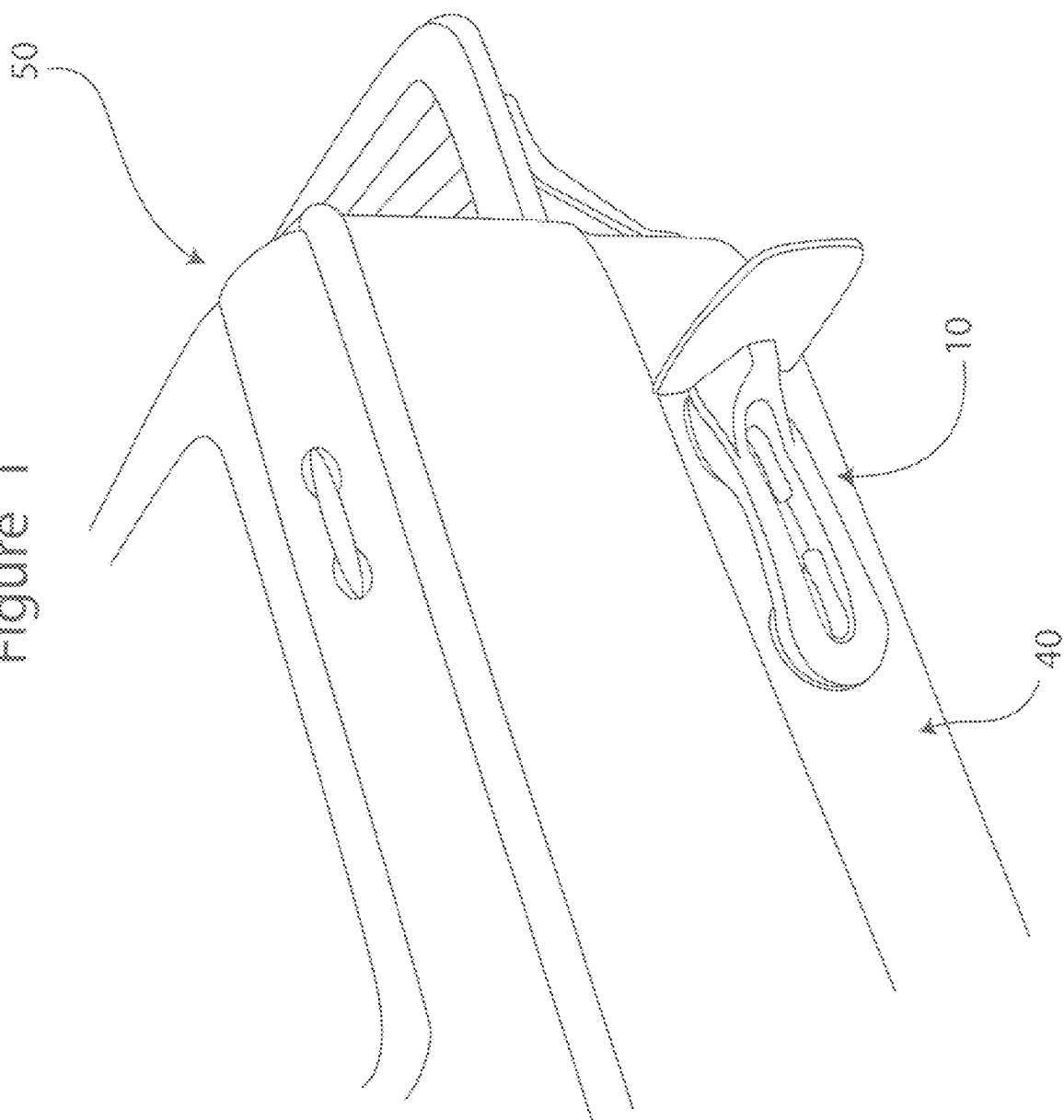
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Figure 1



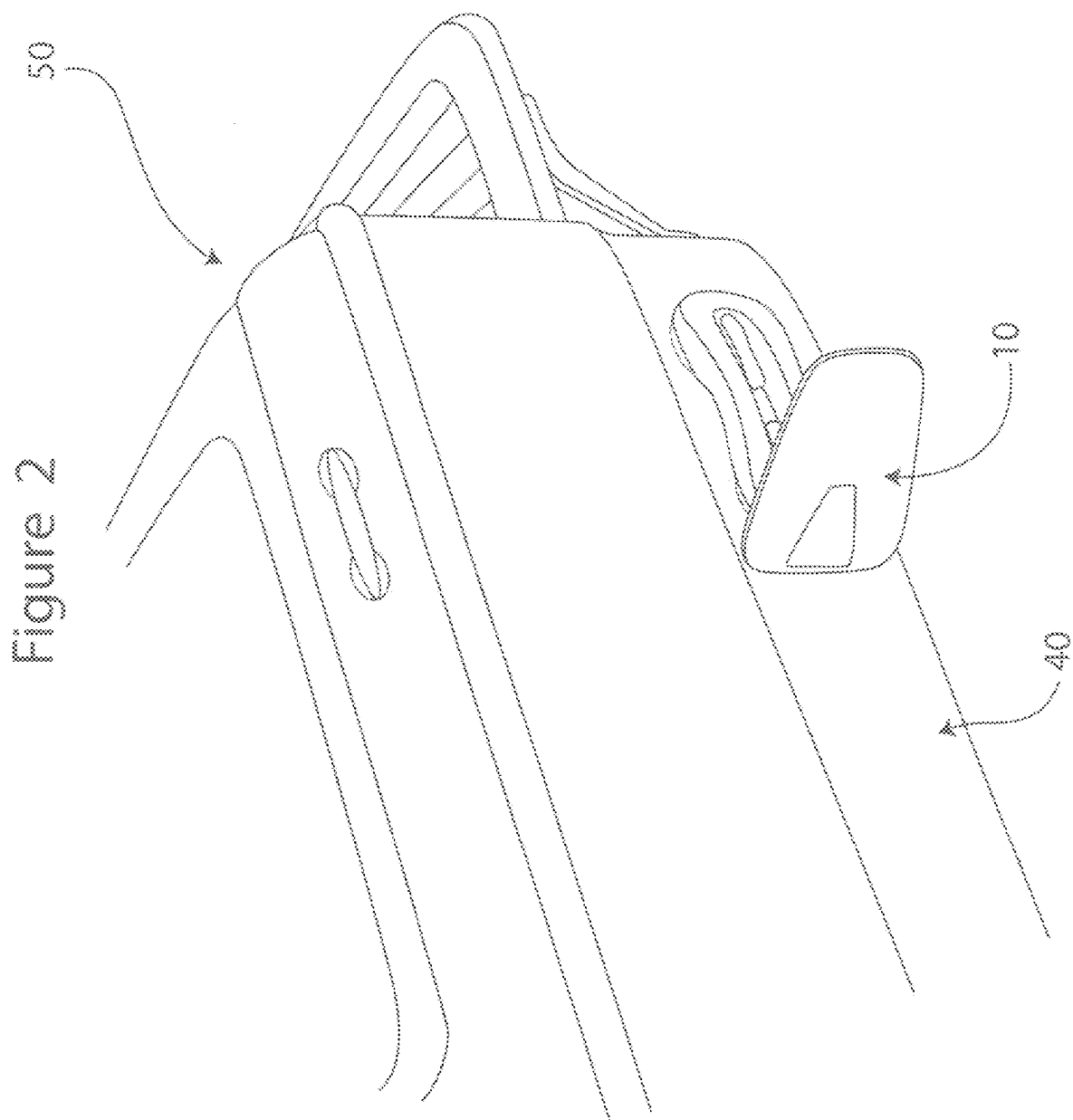


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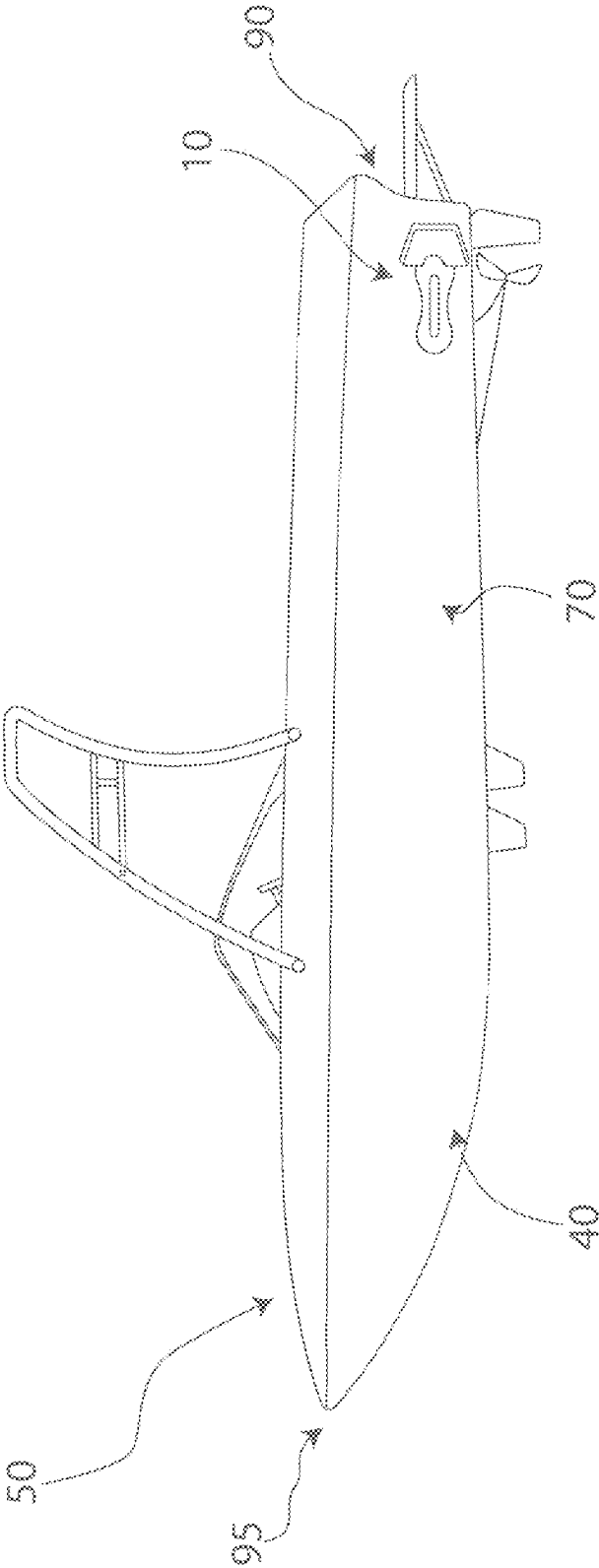


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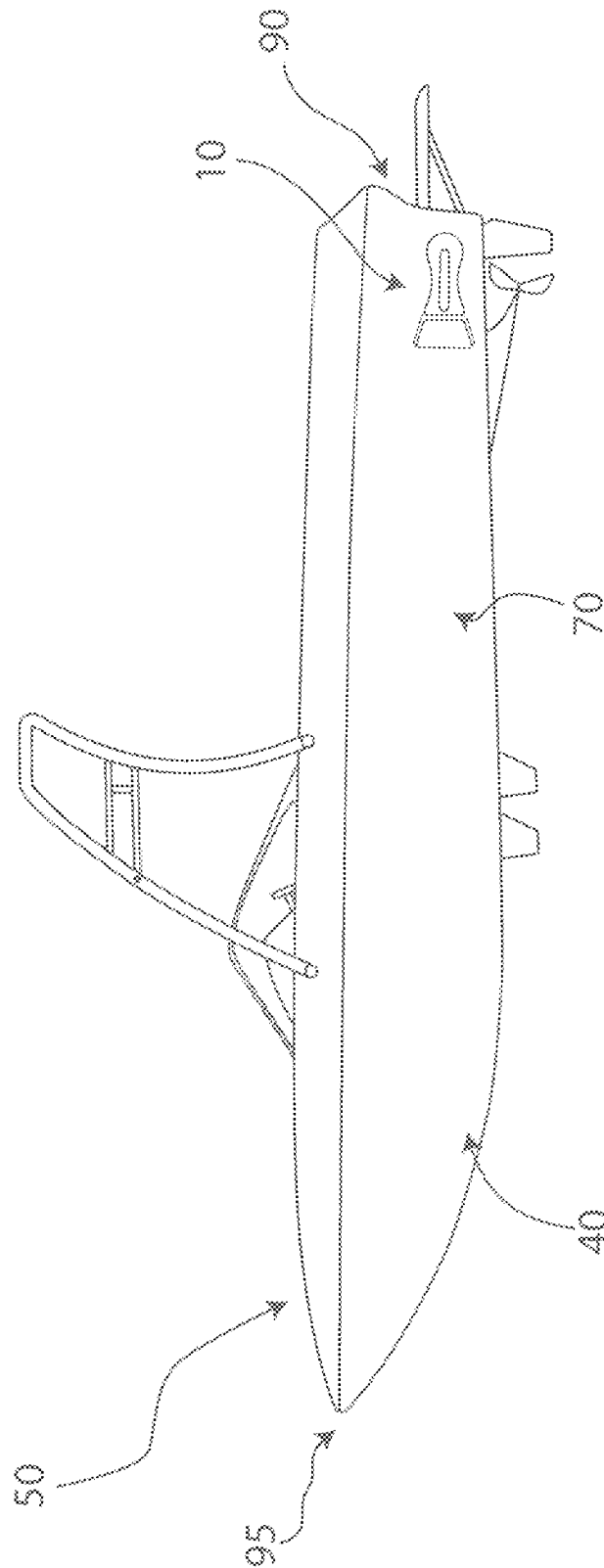


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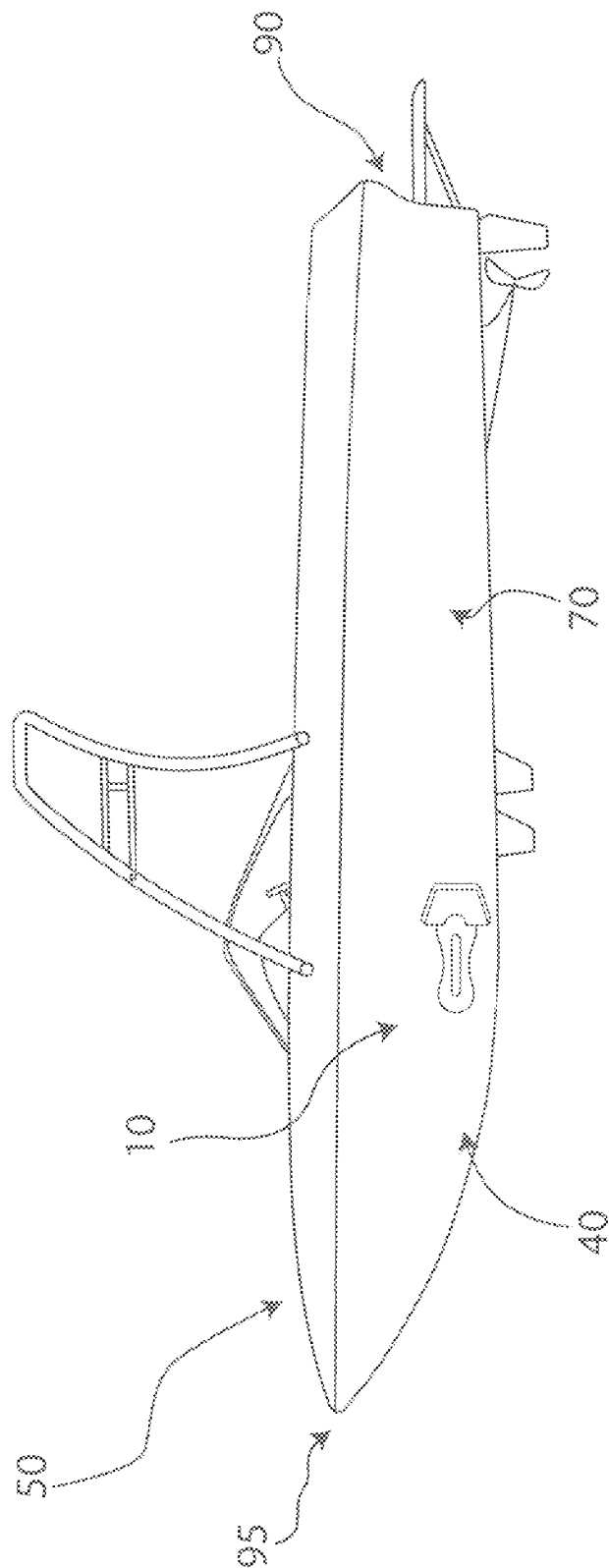
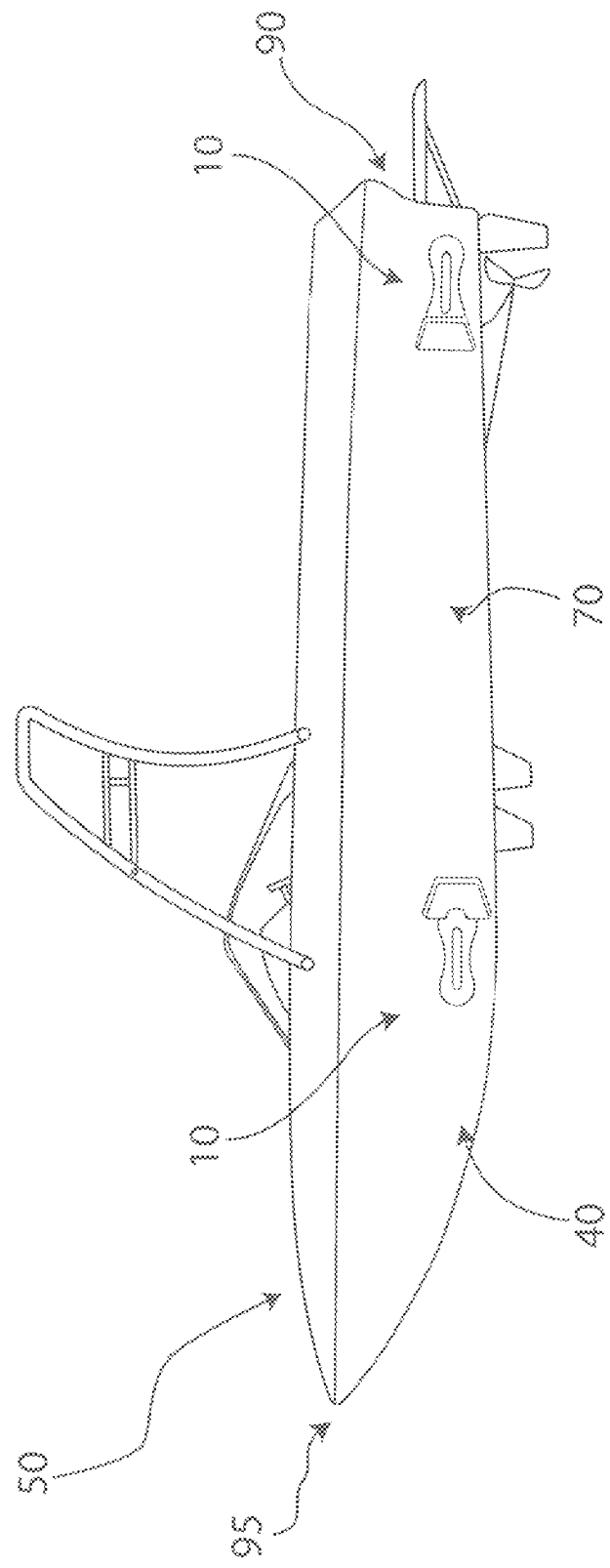
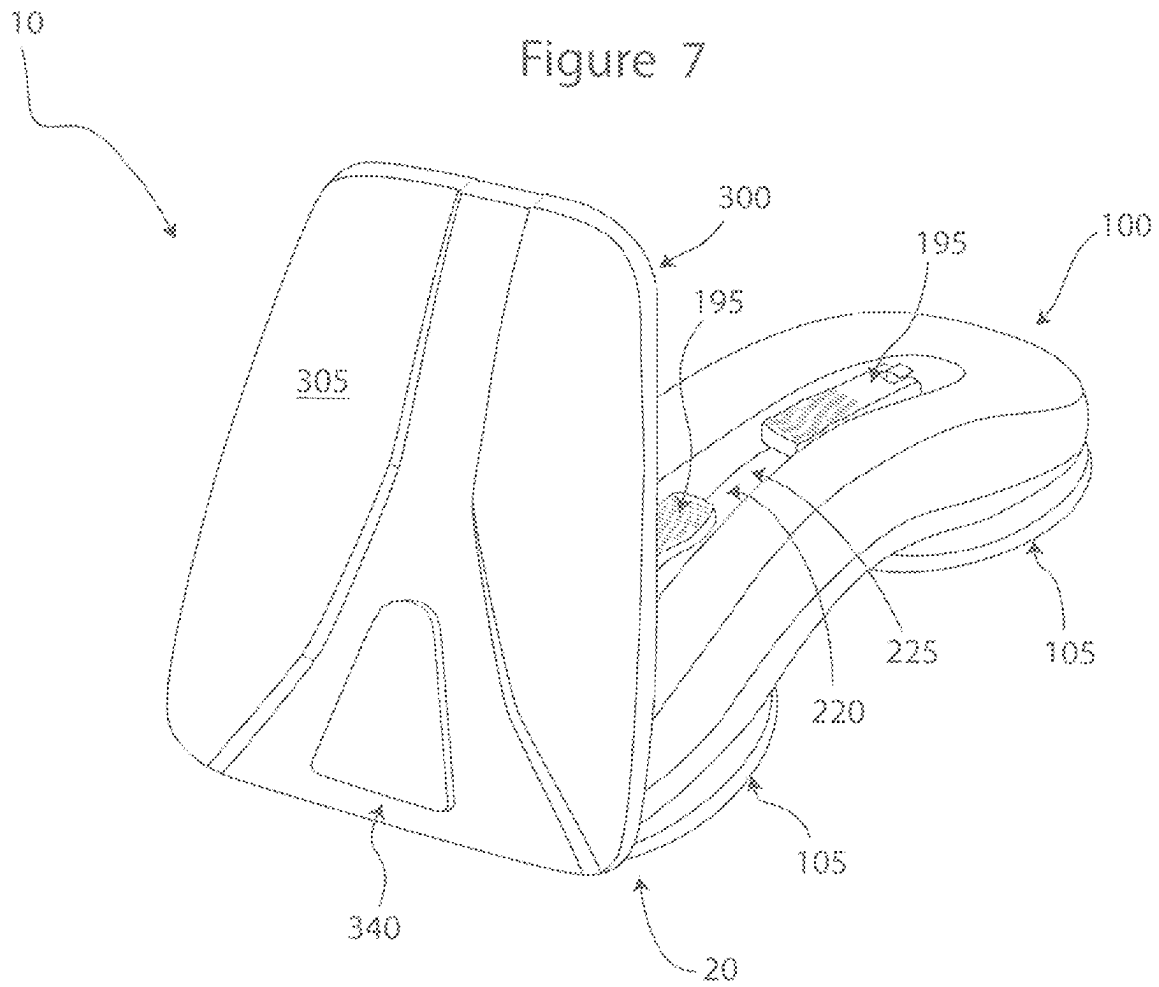
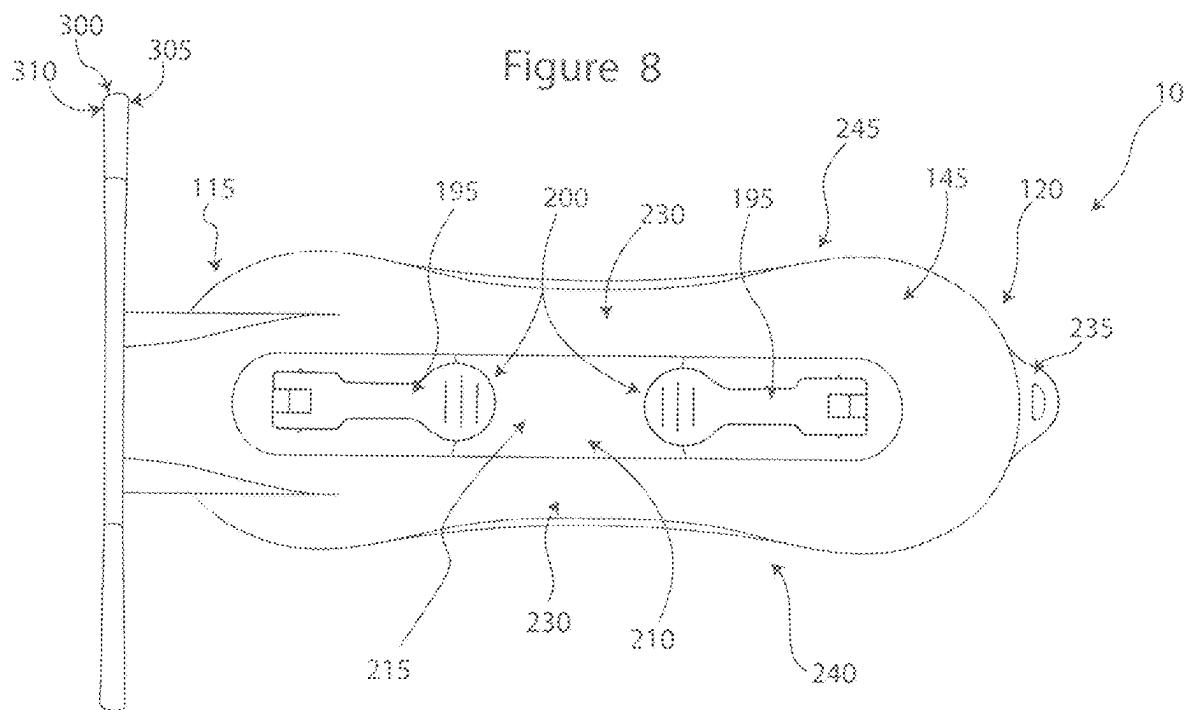


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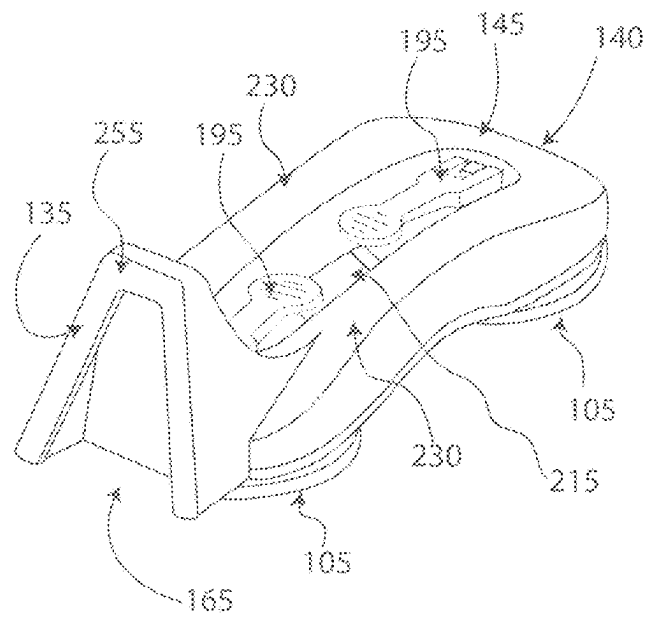
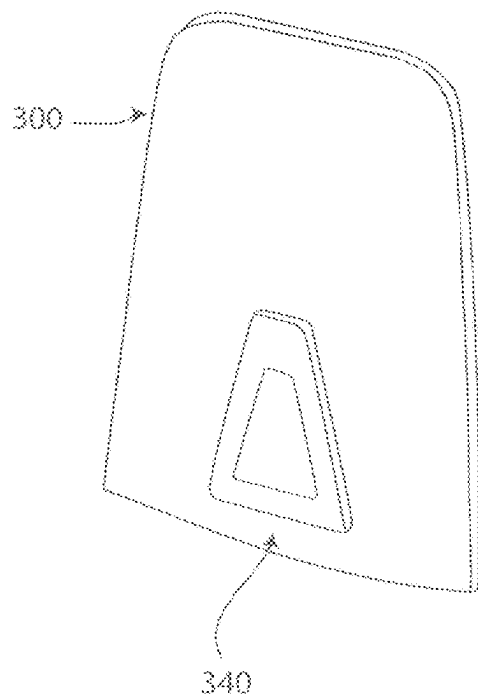


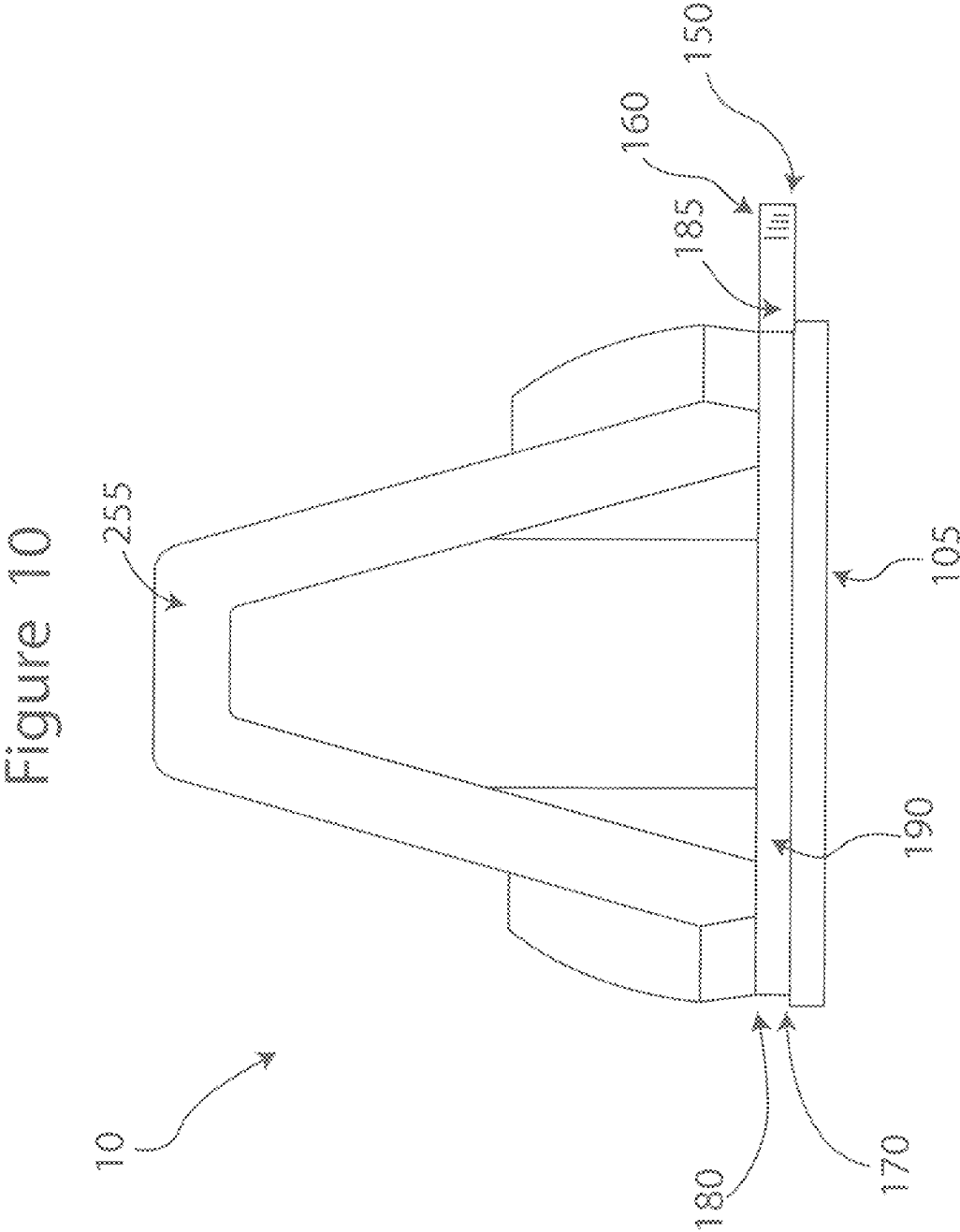


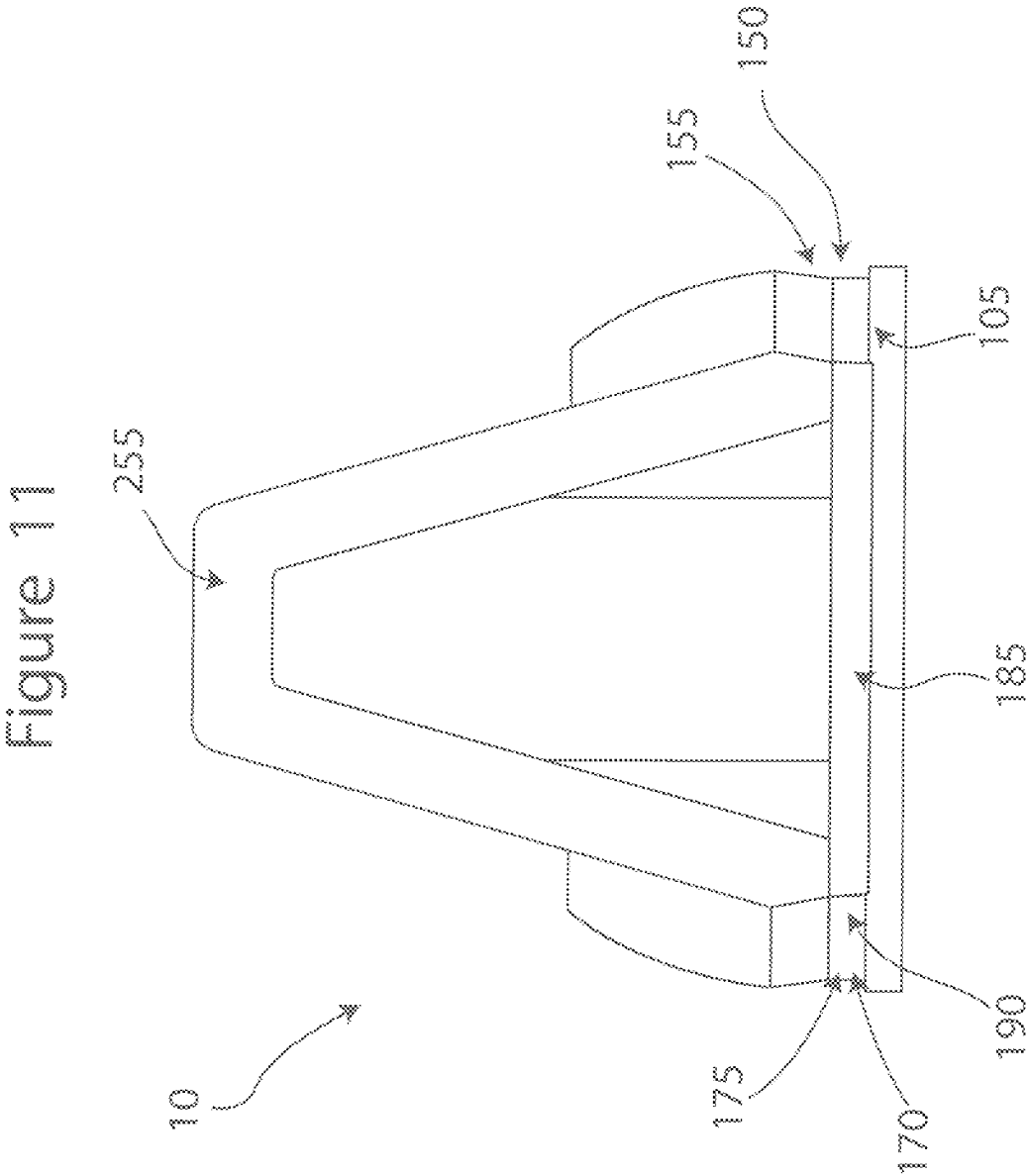
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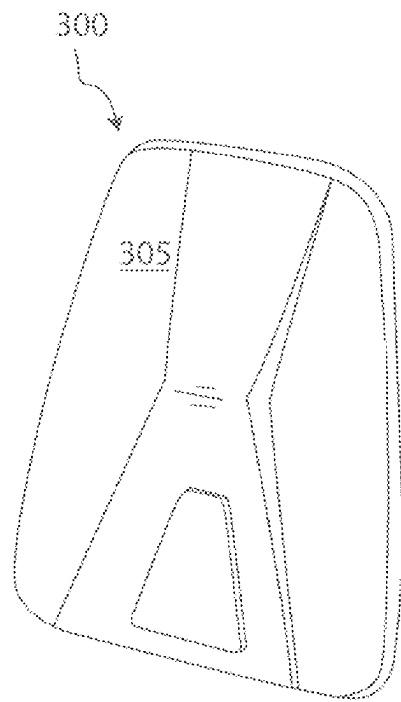


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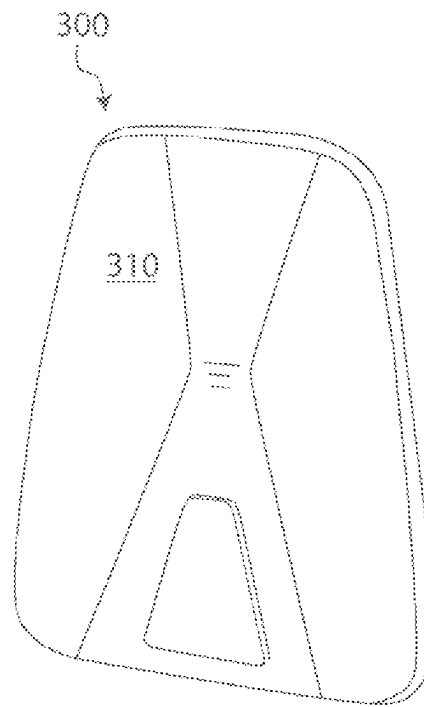


Figure 13

Figure 14

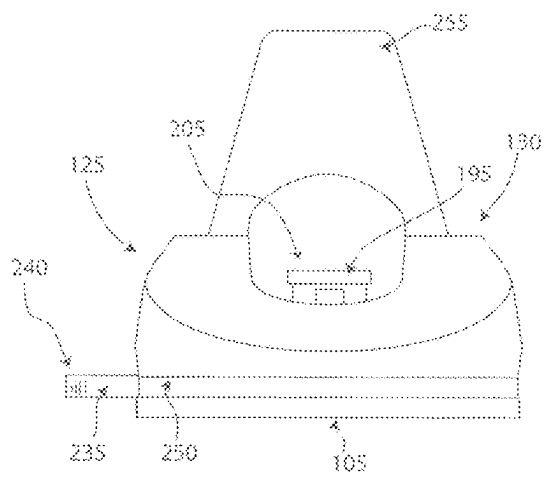
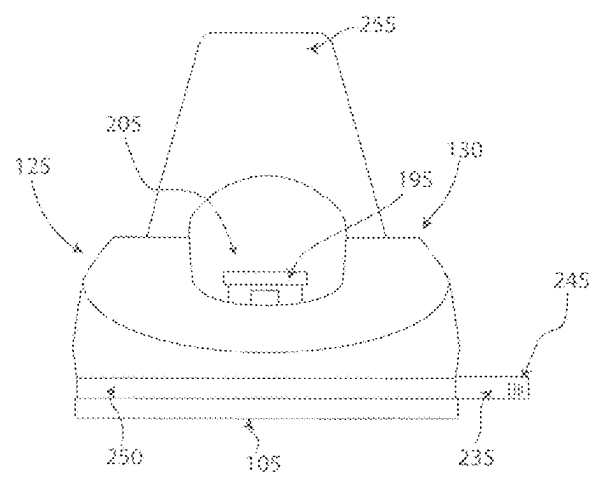


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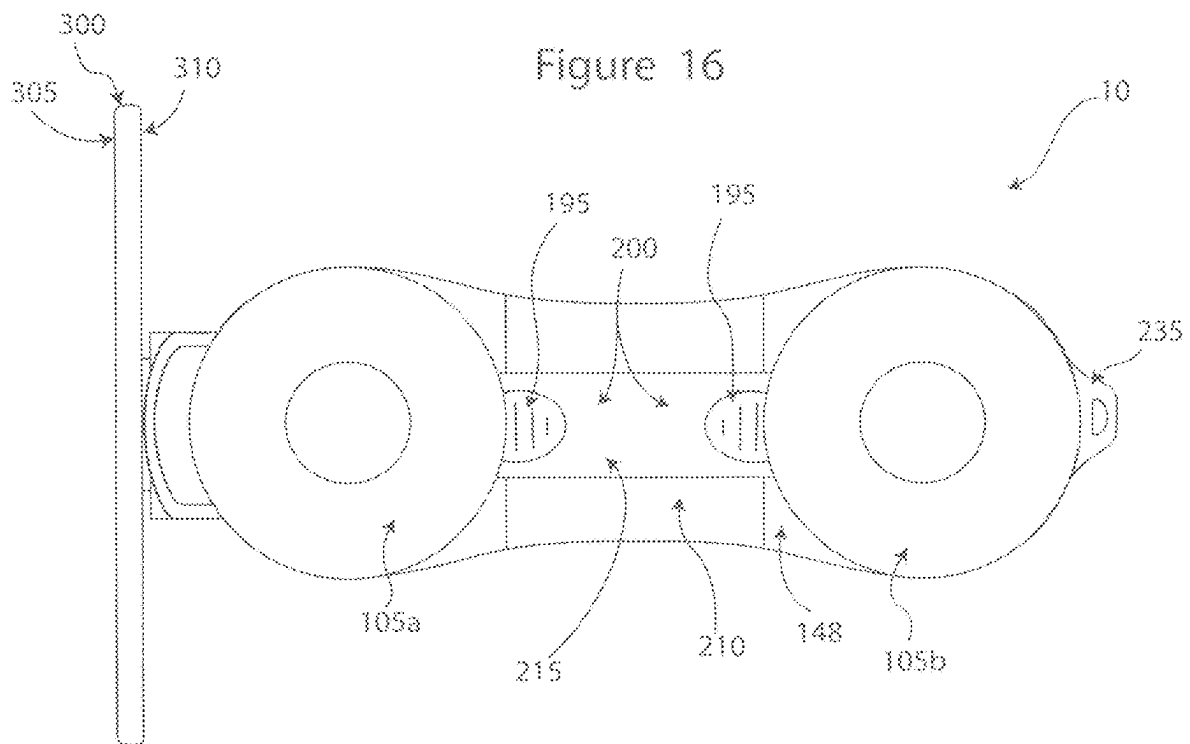
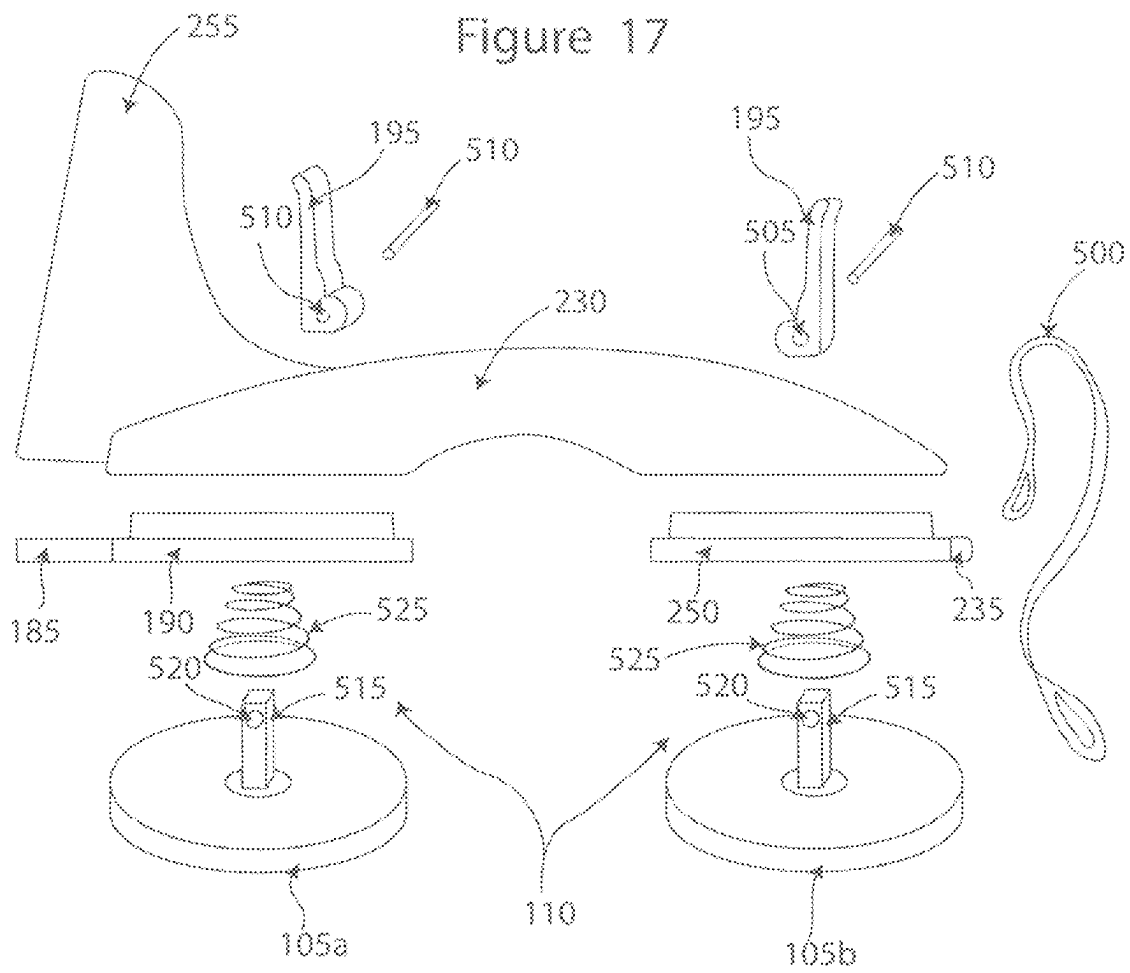


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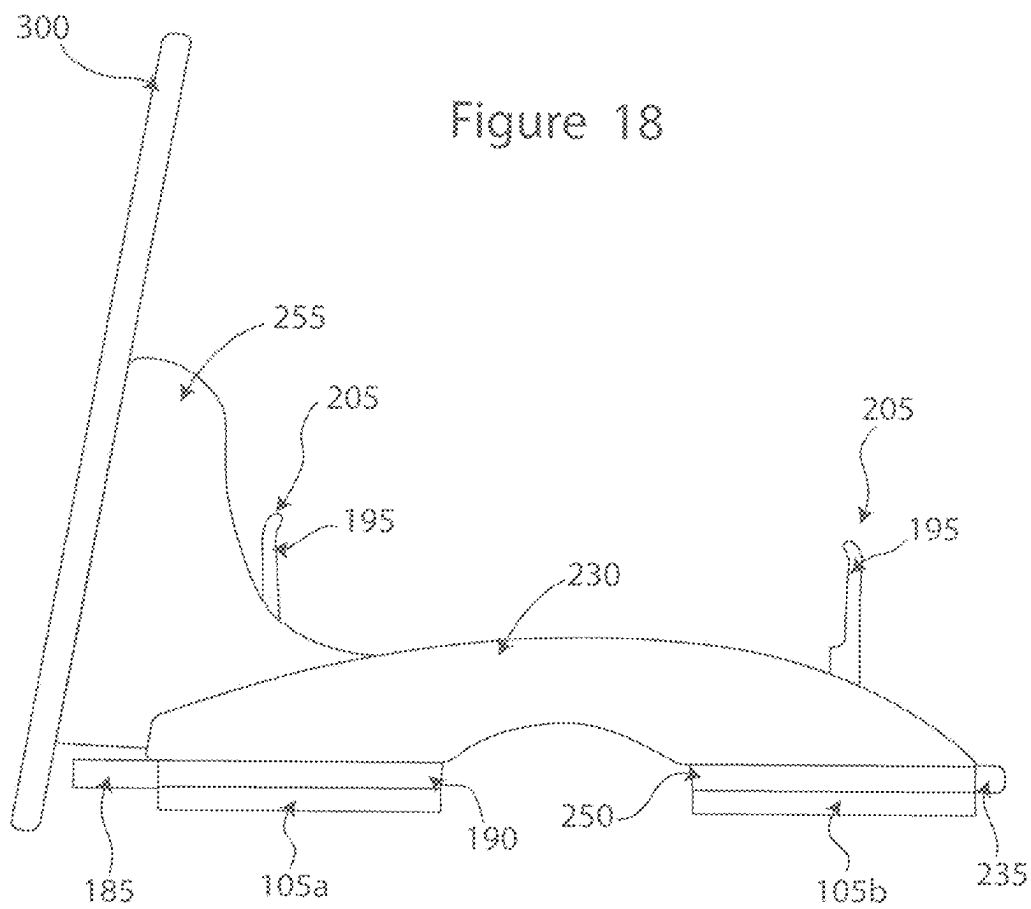
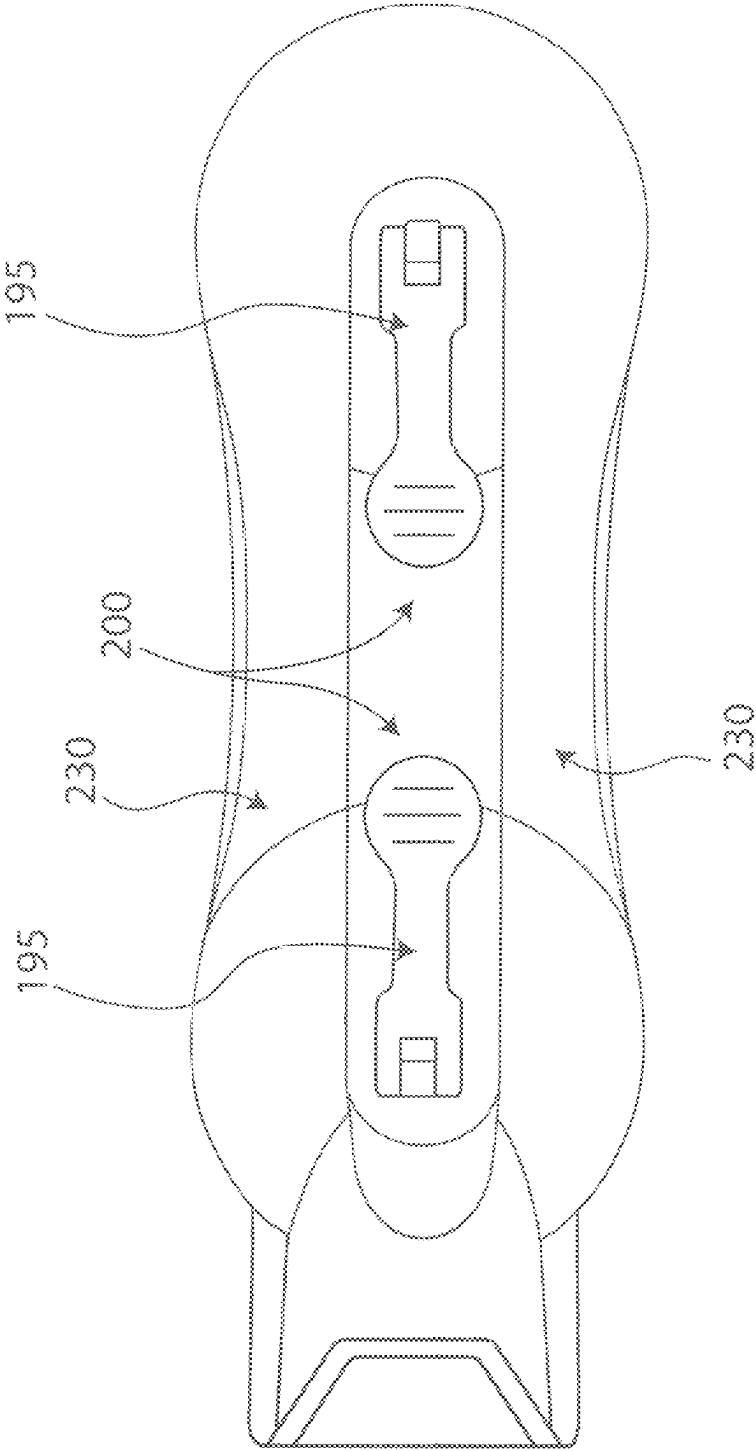
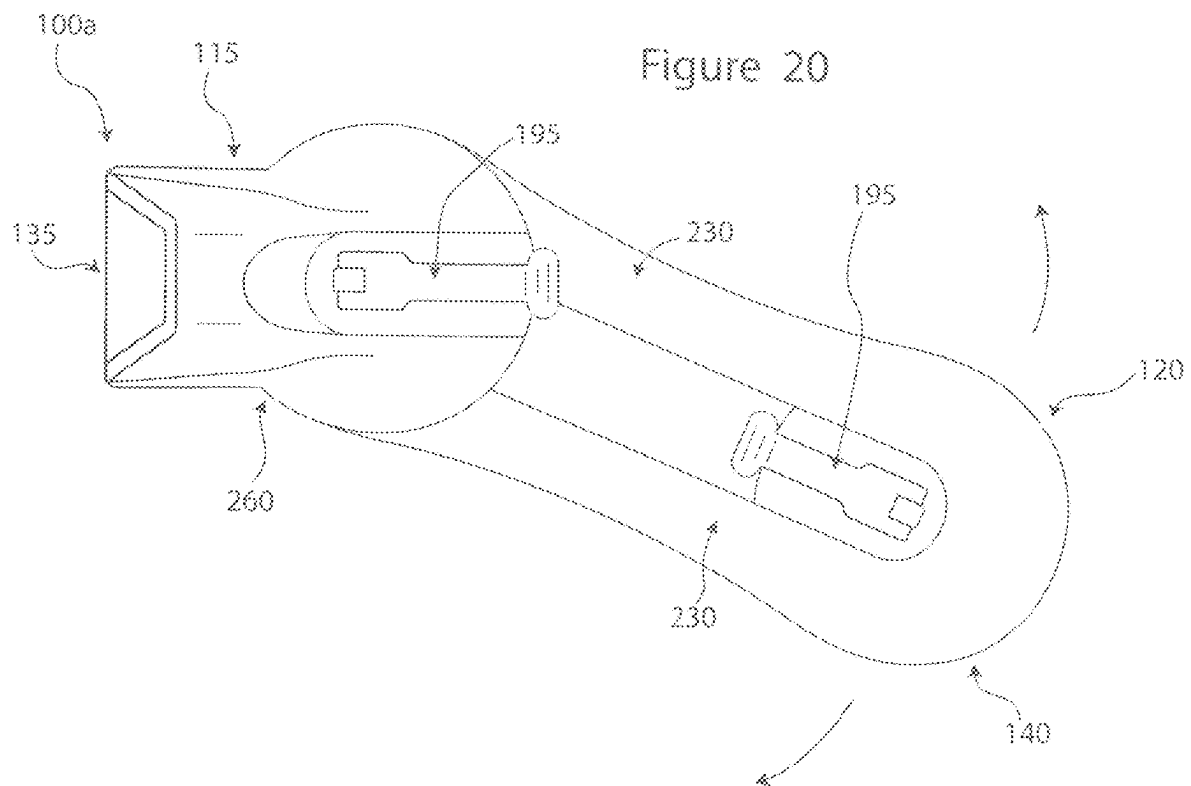


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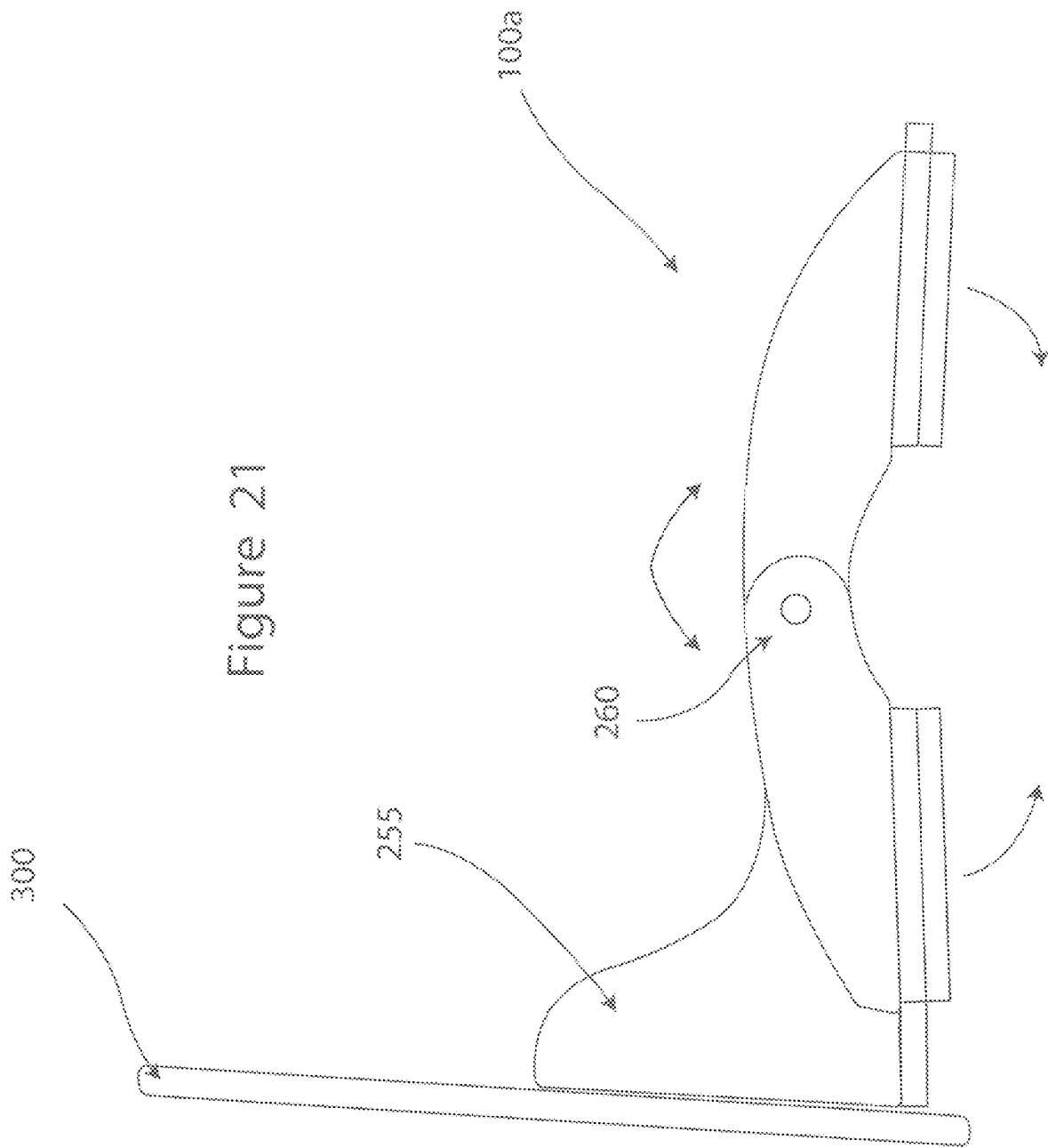


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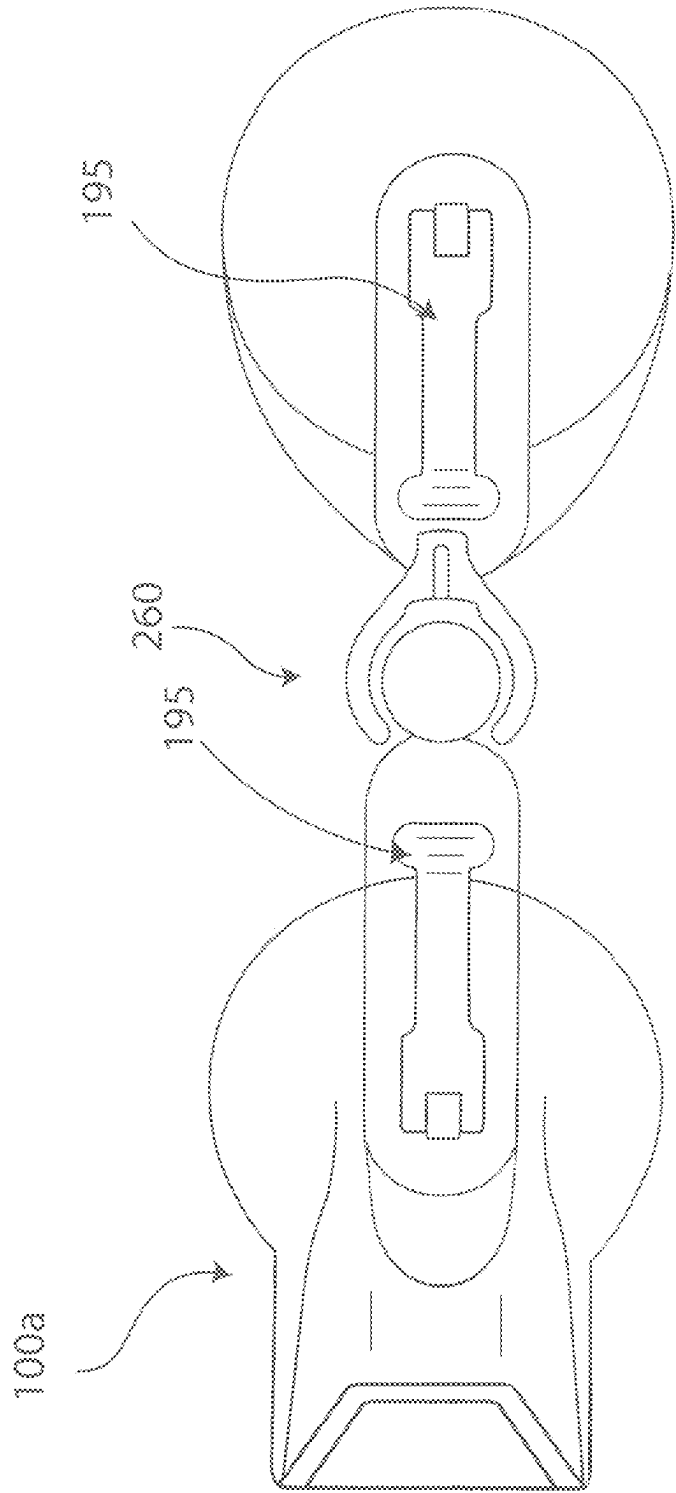


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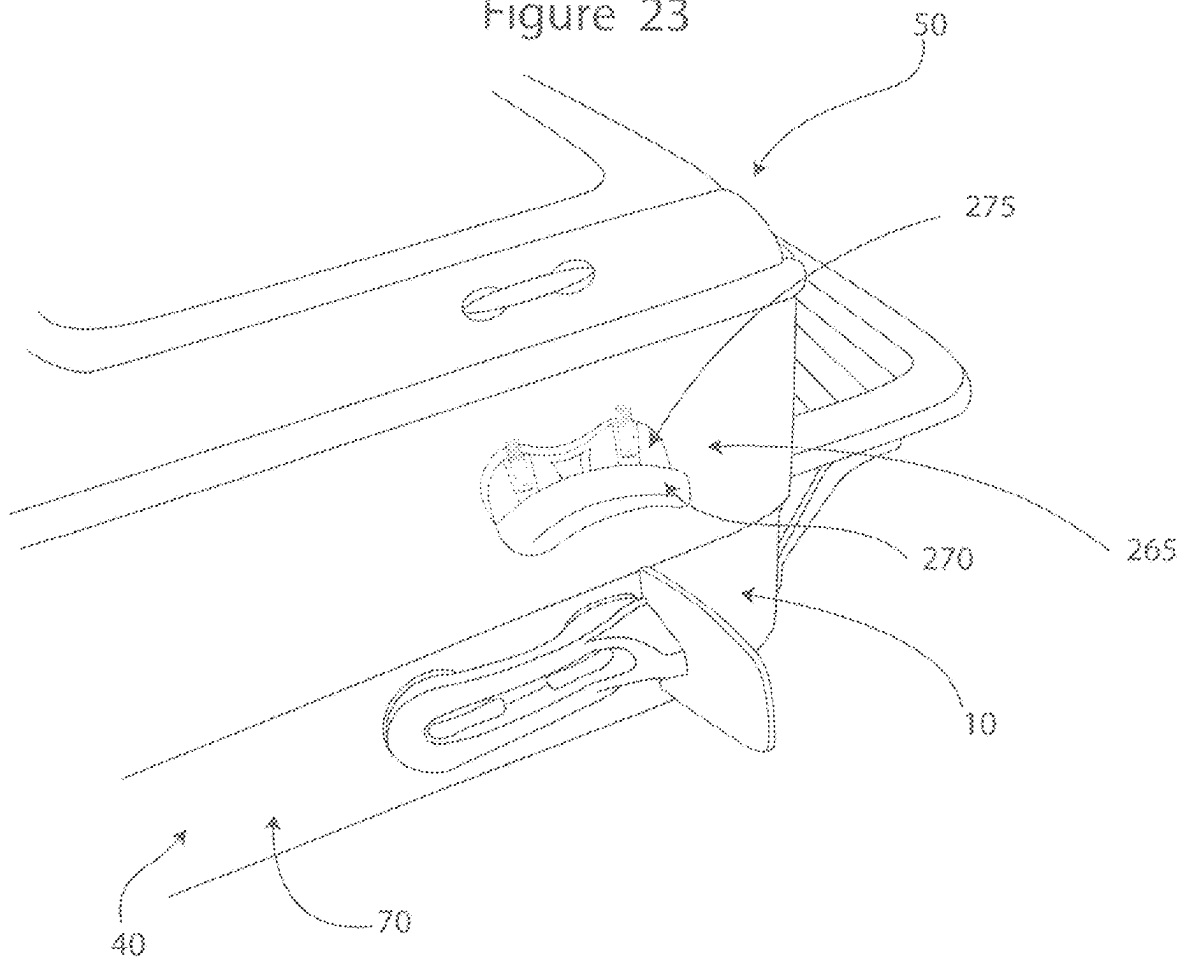


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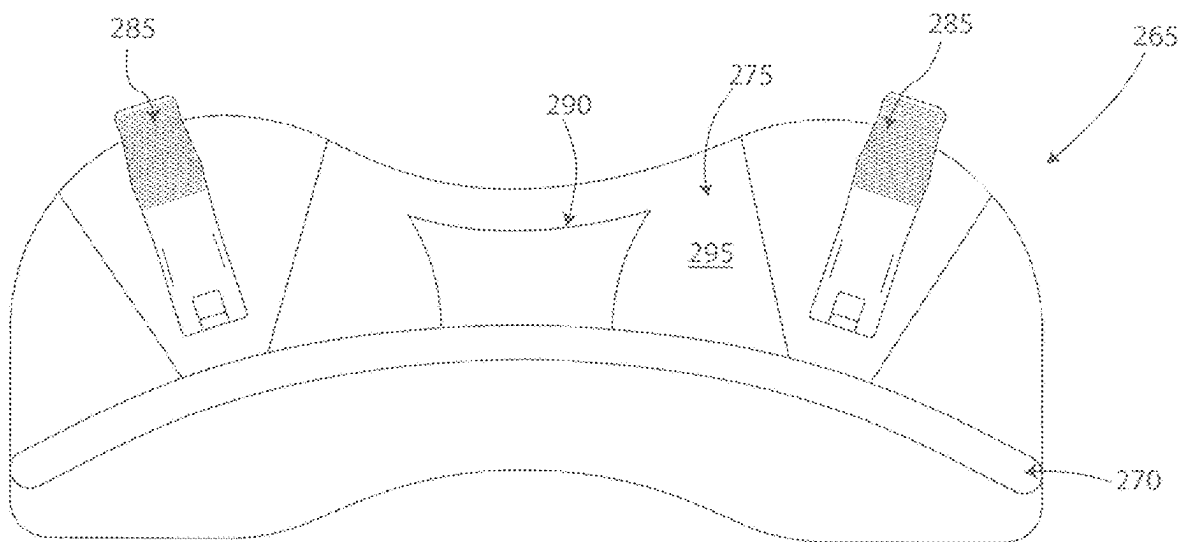


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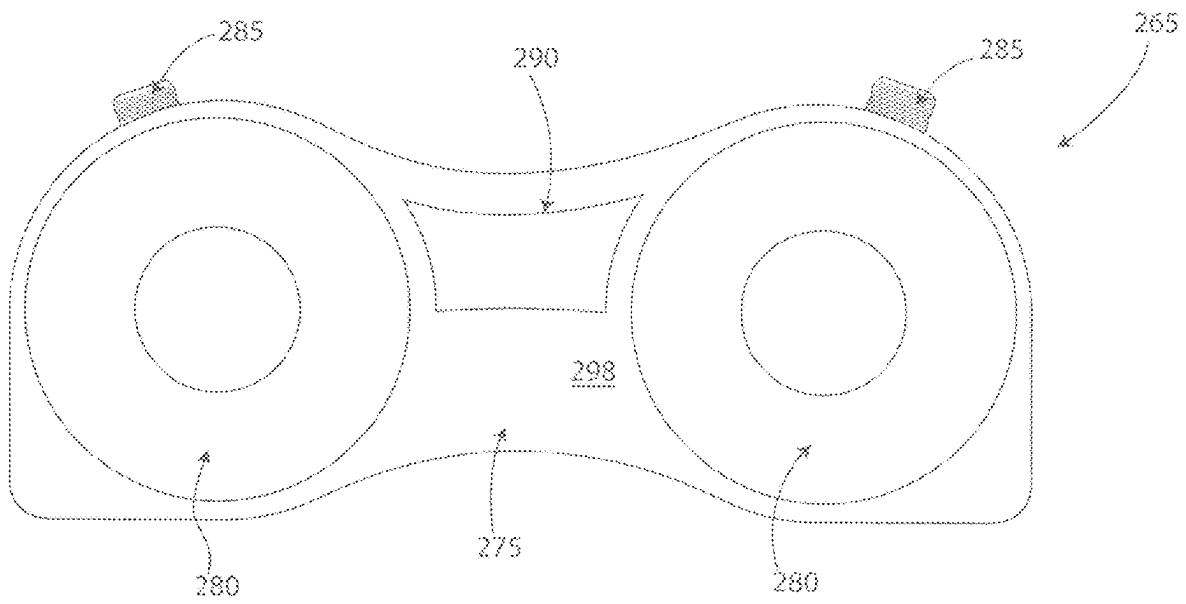


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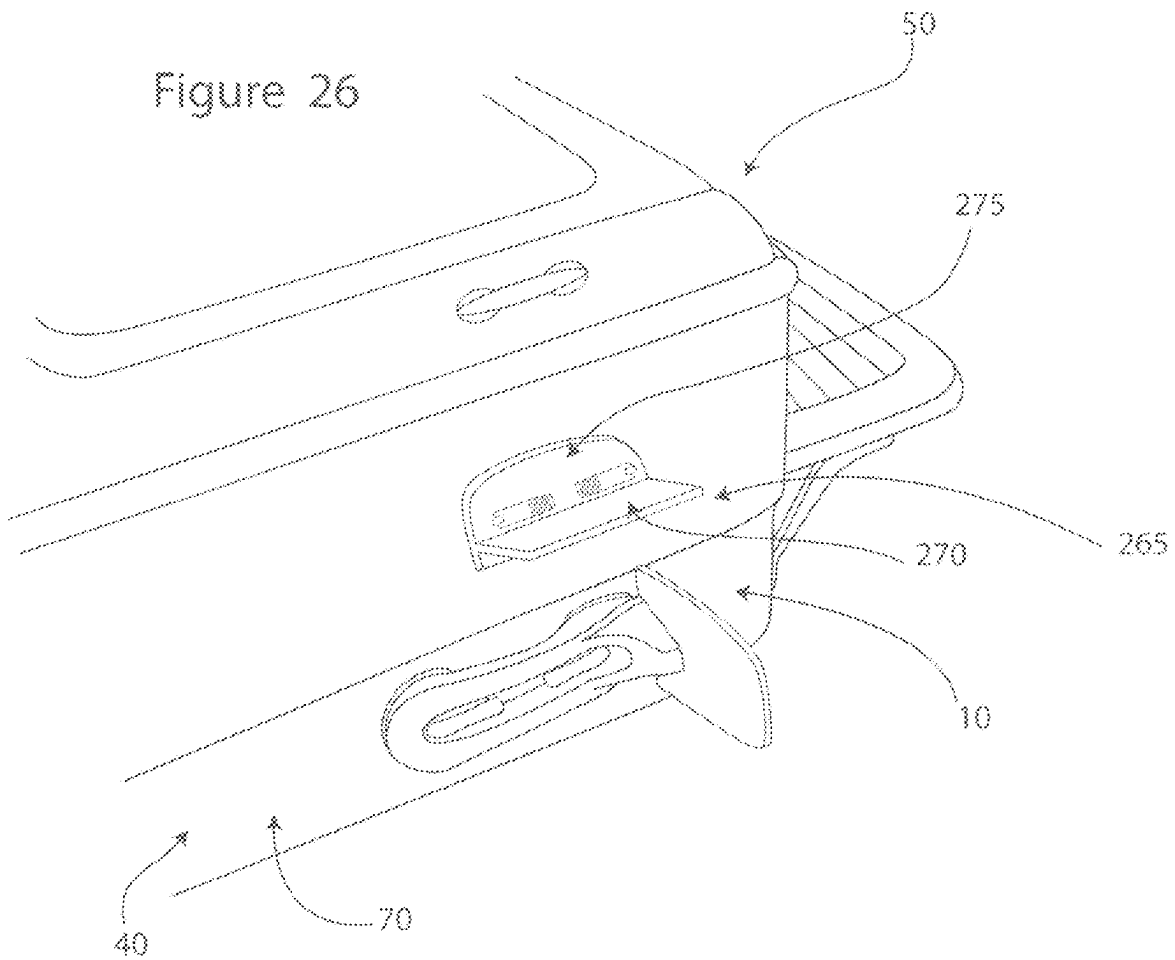


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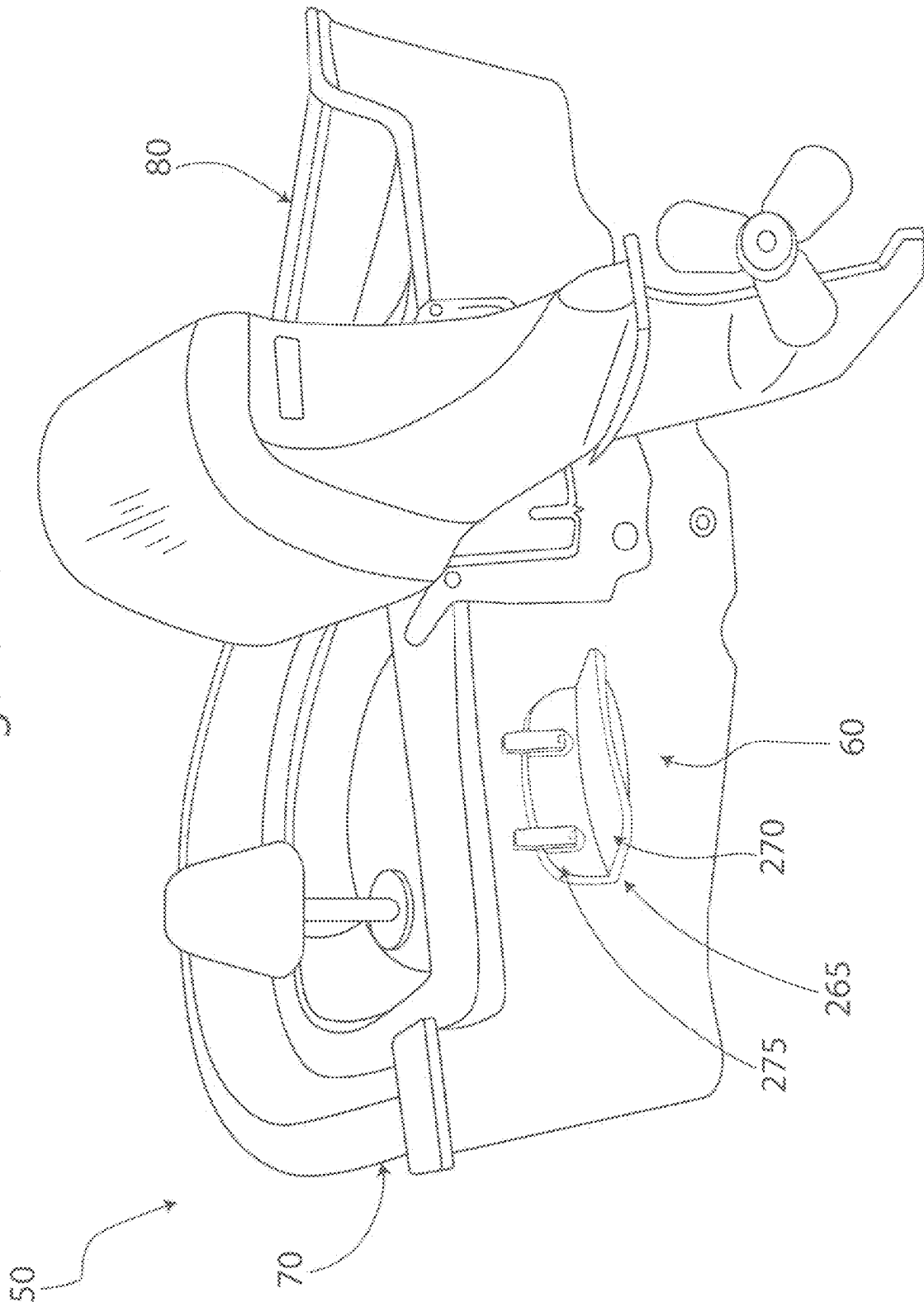
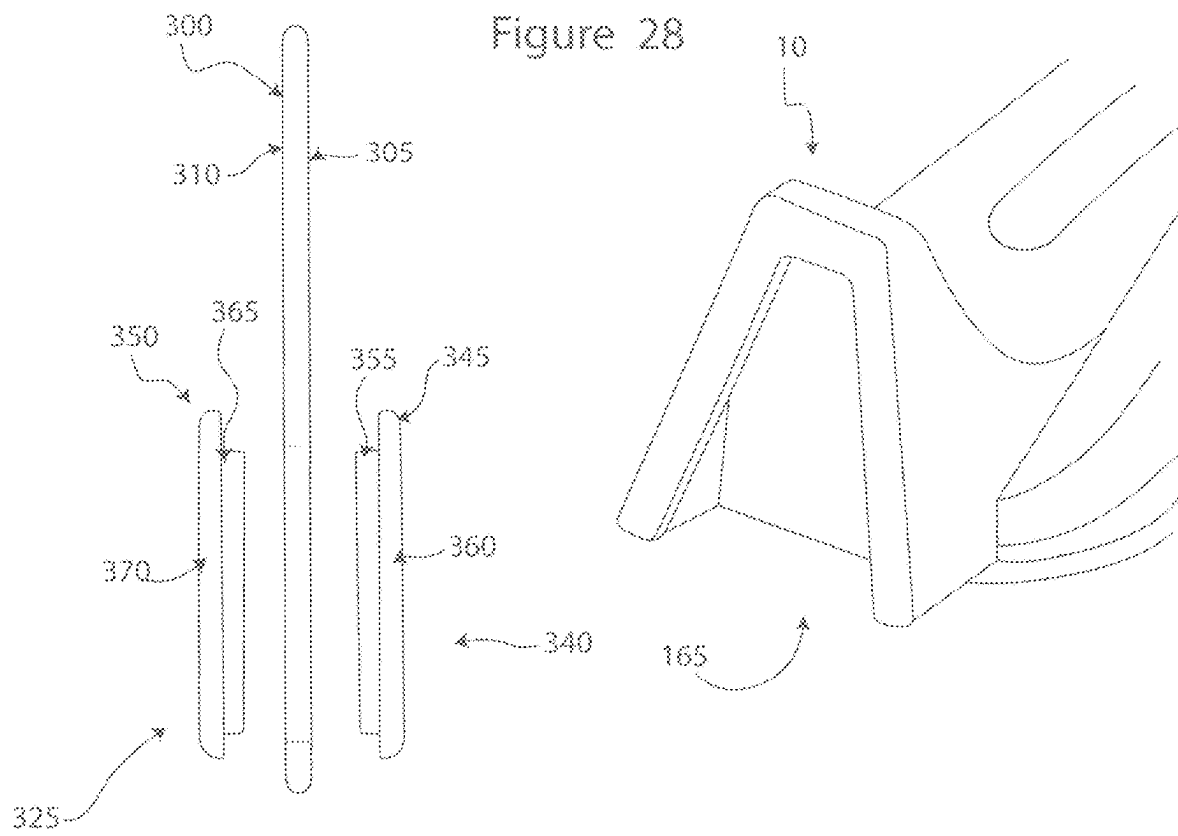


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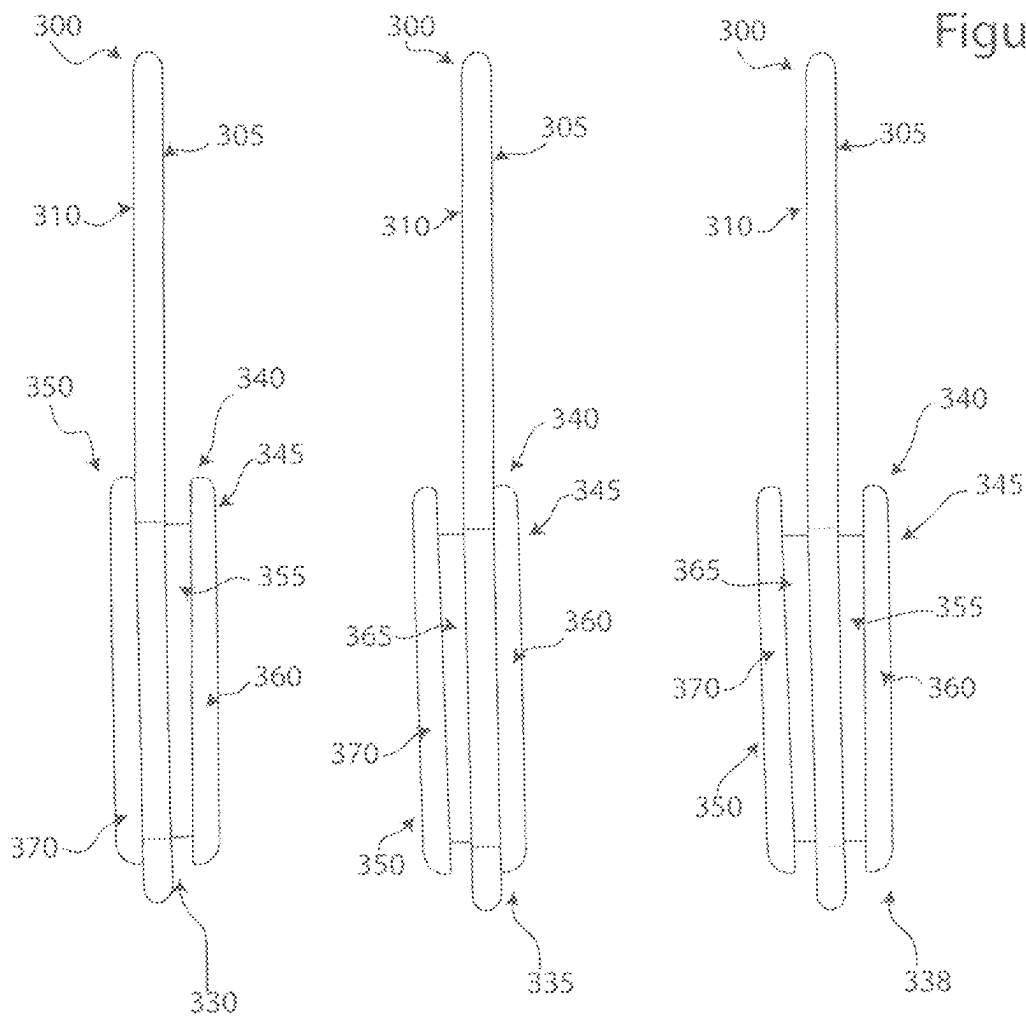


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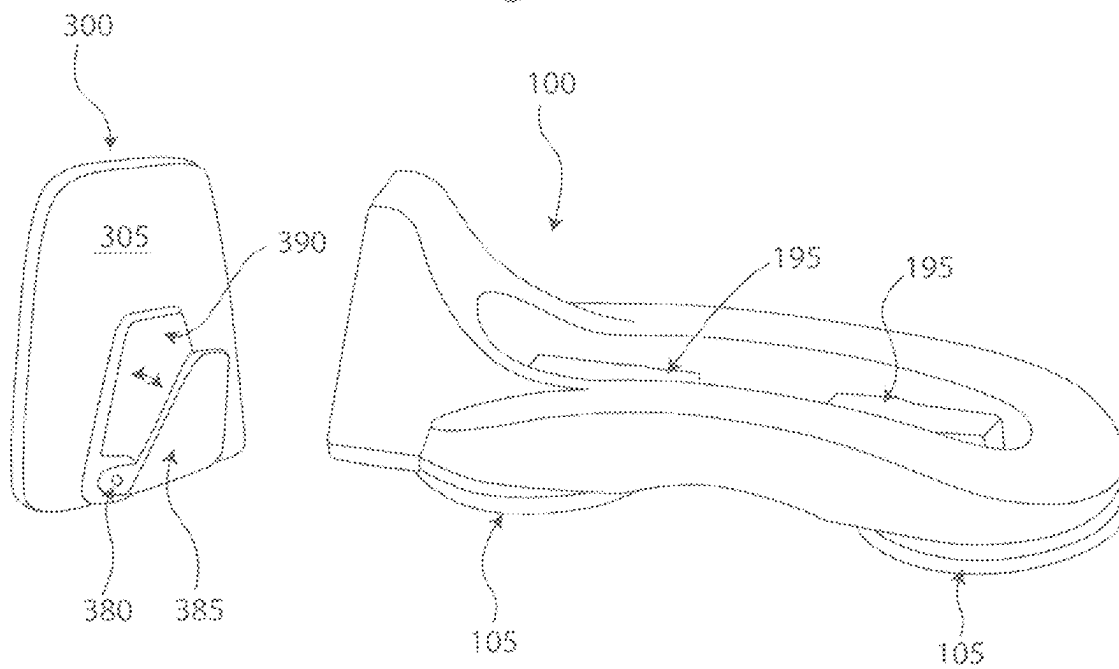


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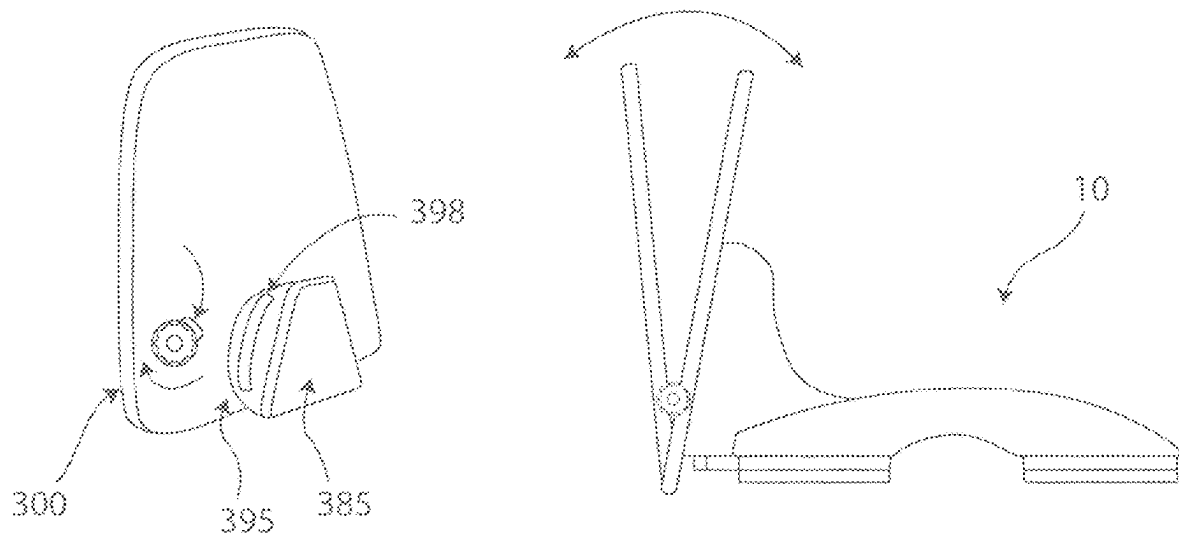


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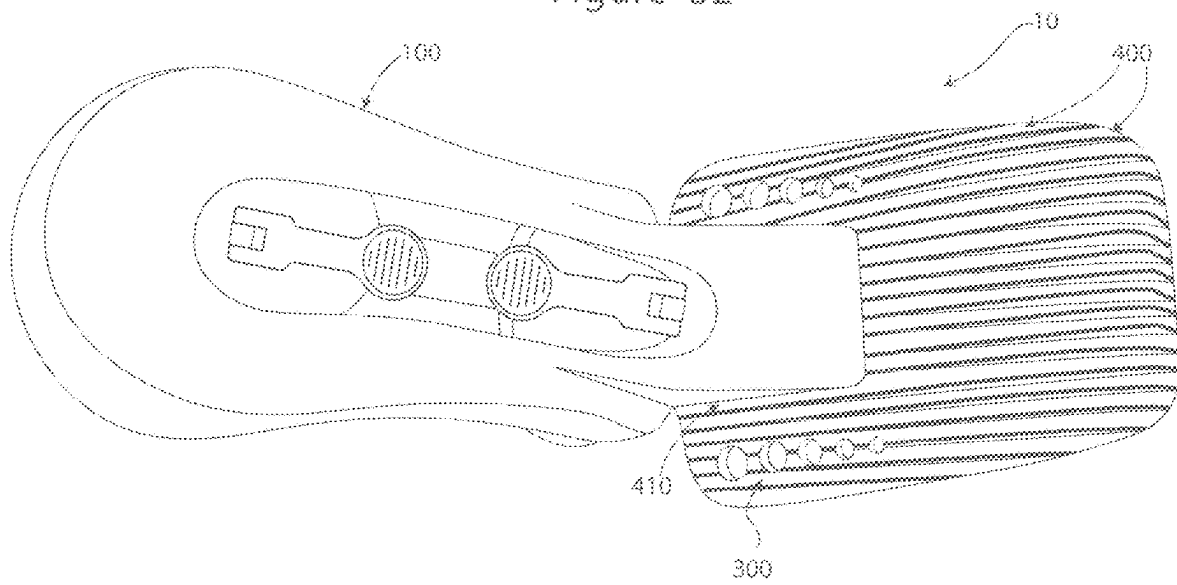
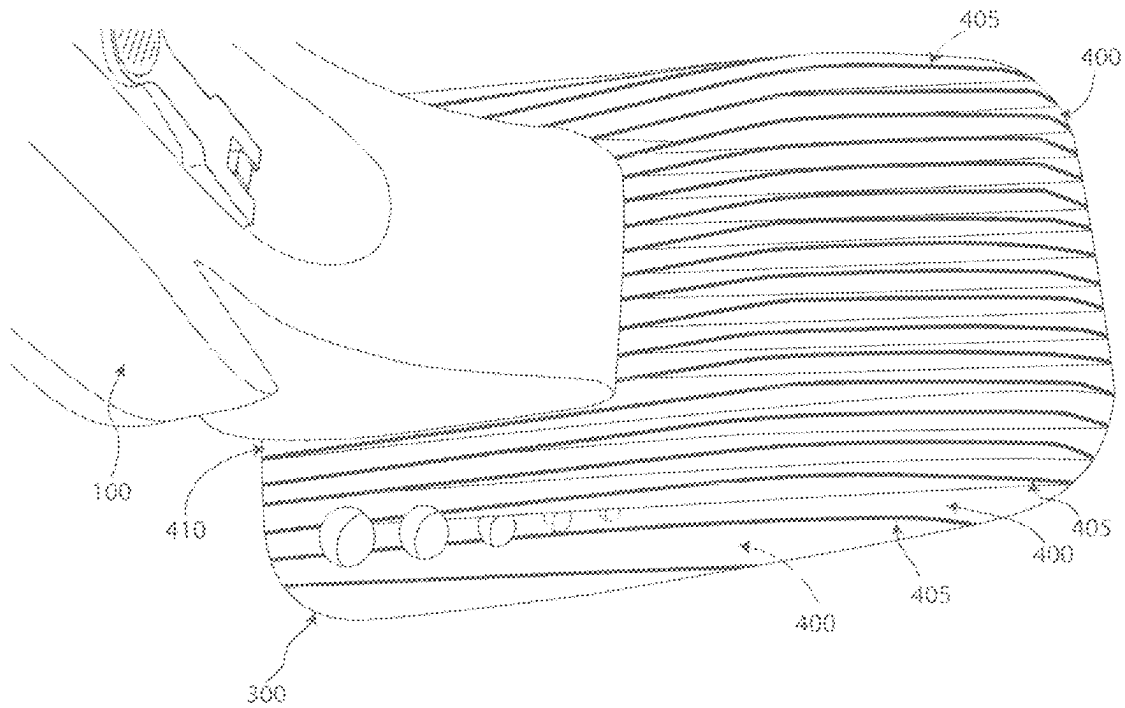


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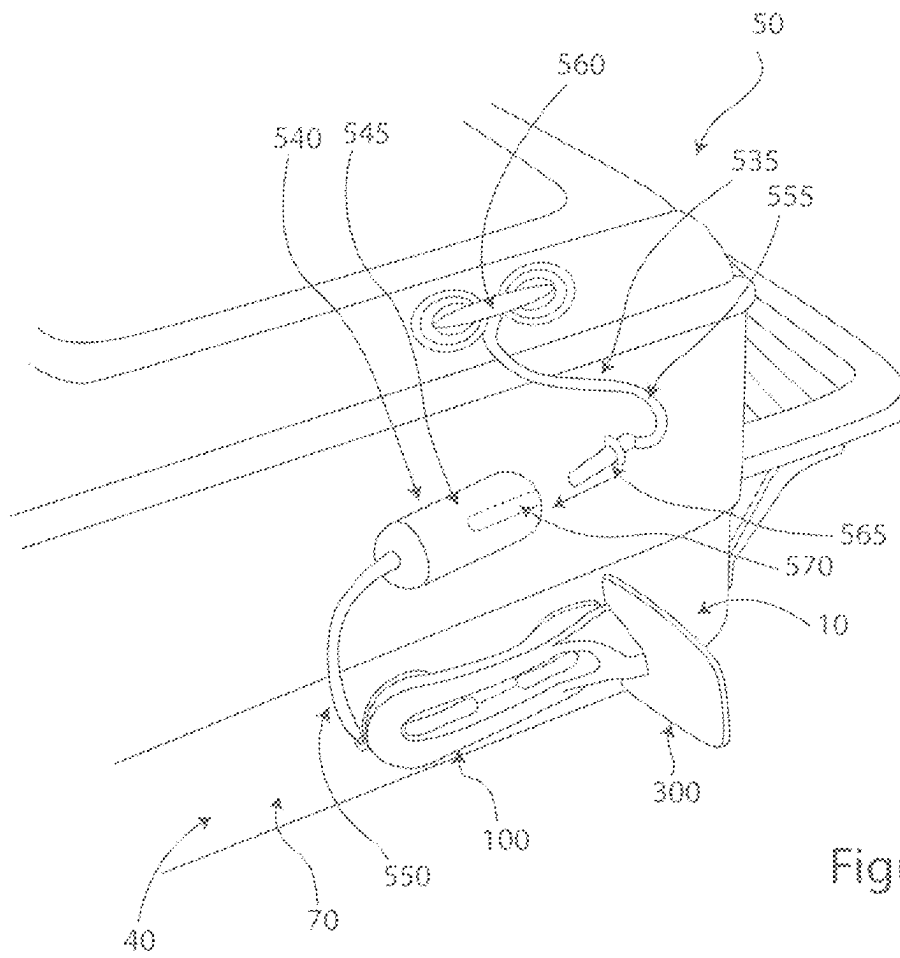


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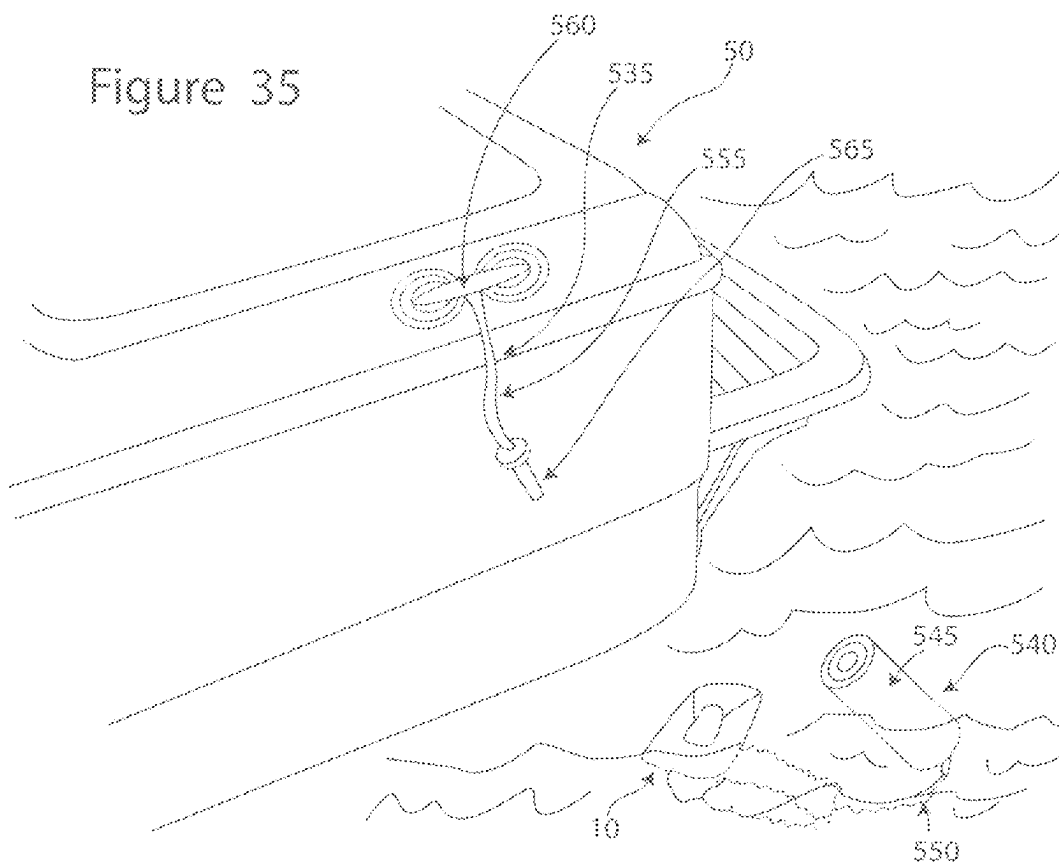


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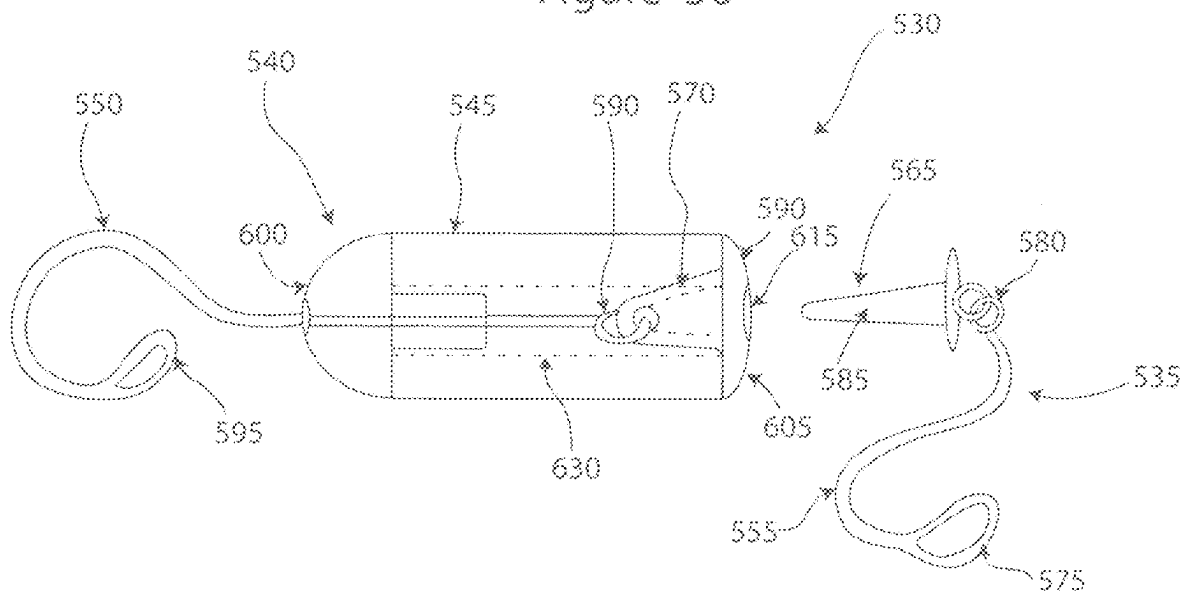
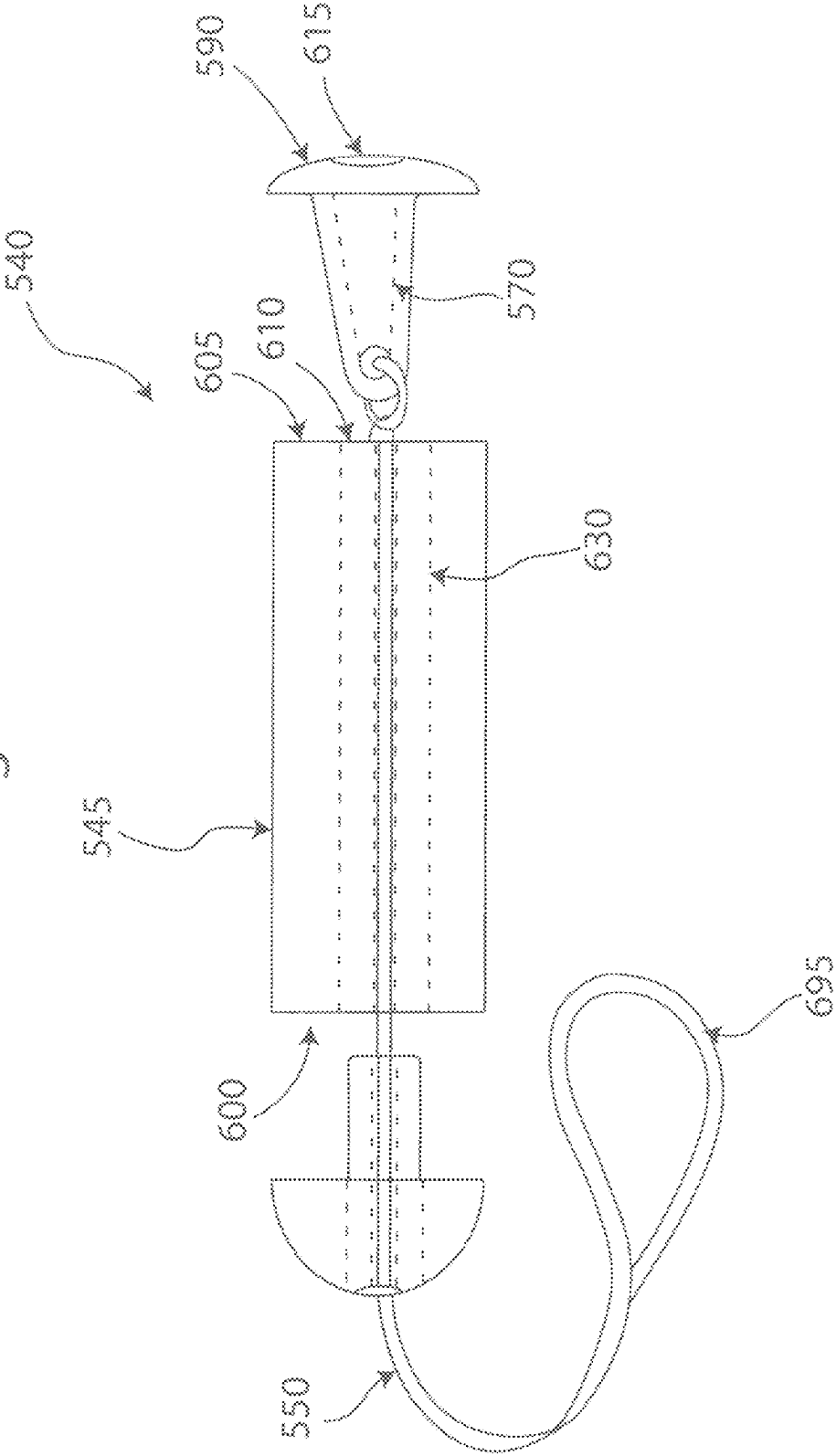


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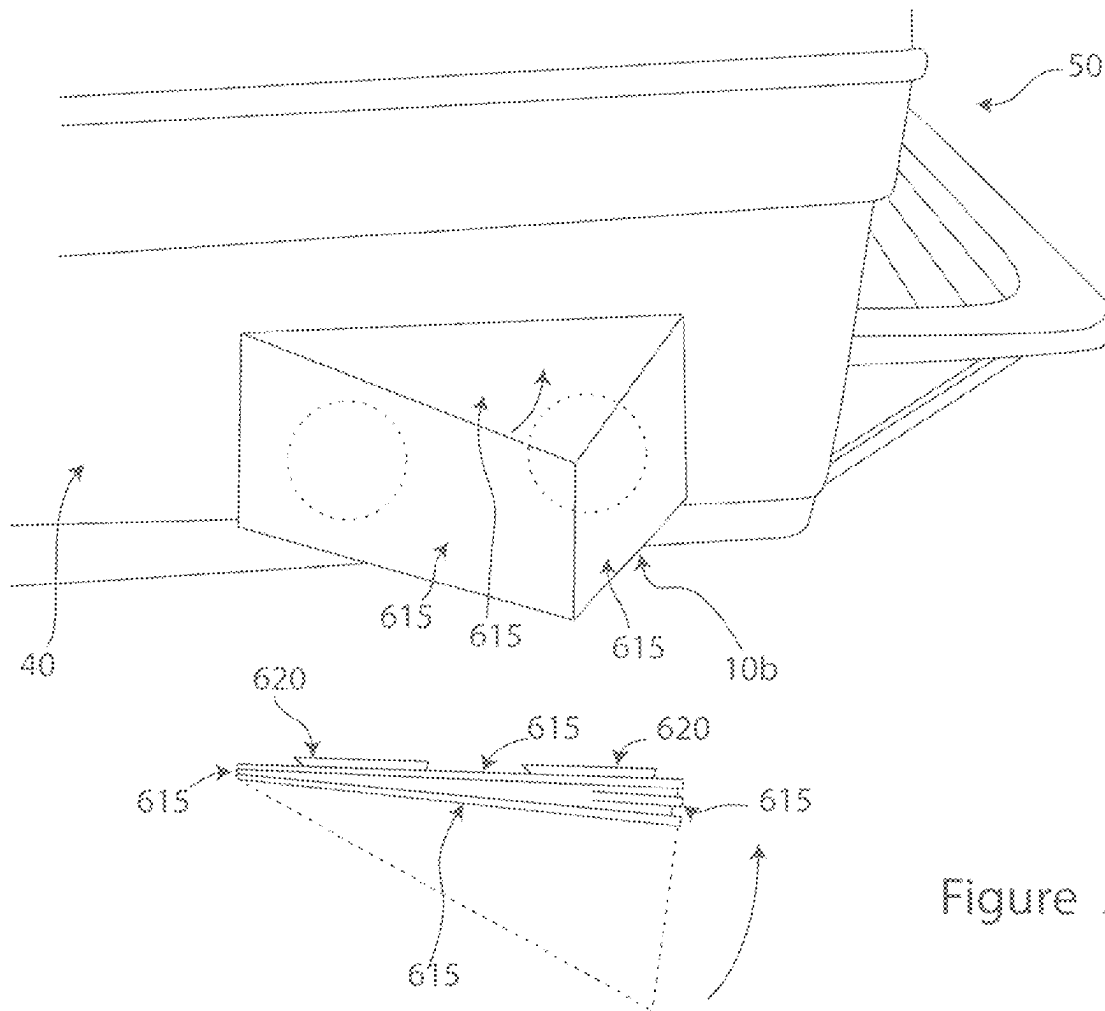


Figure 38

Figure 39

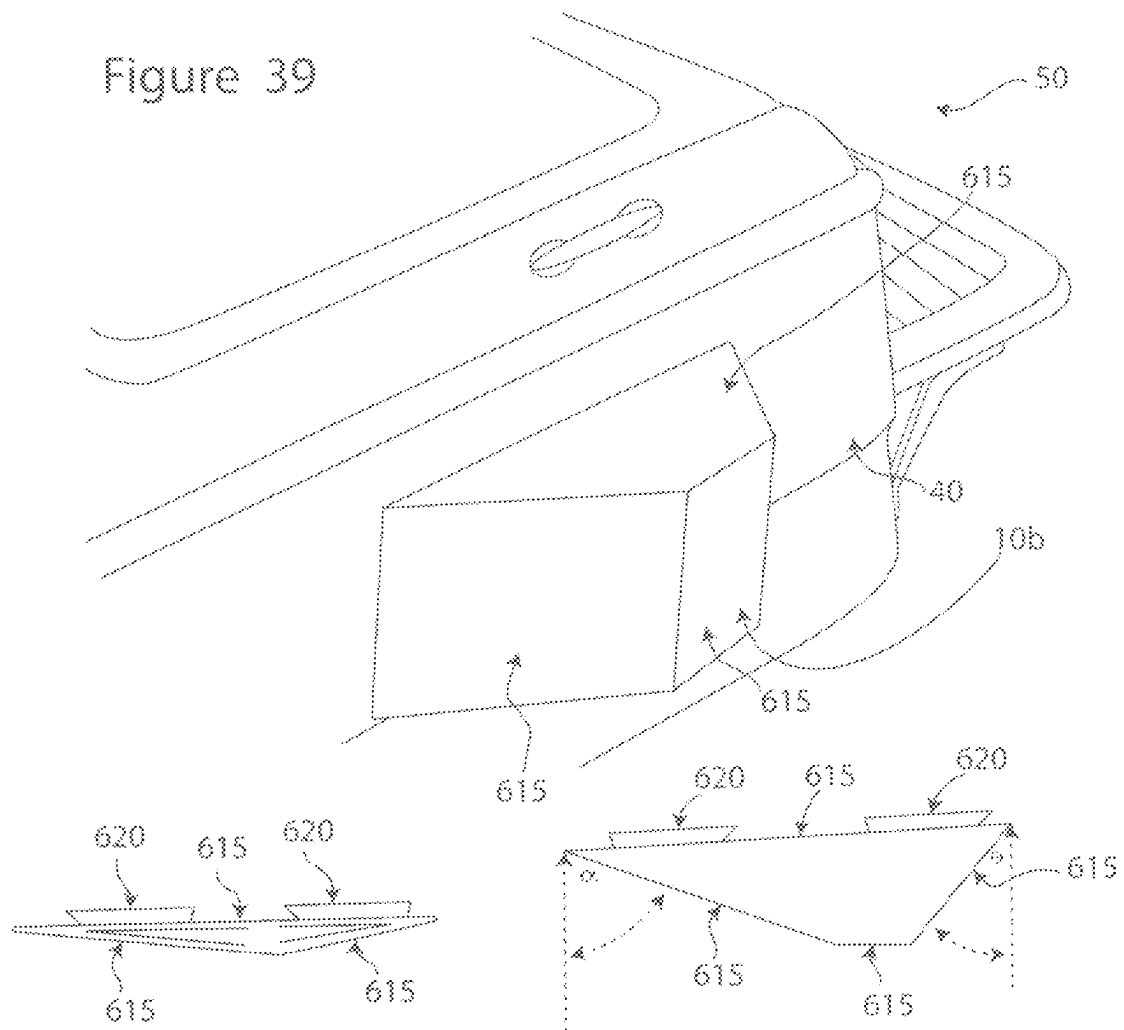


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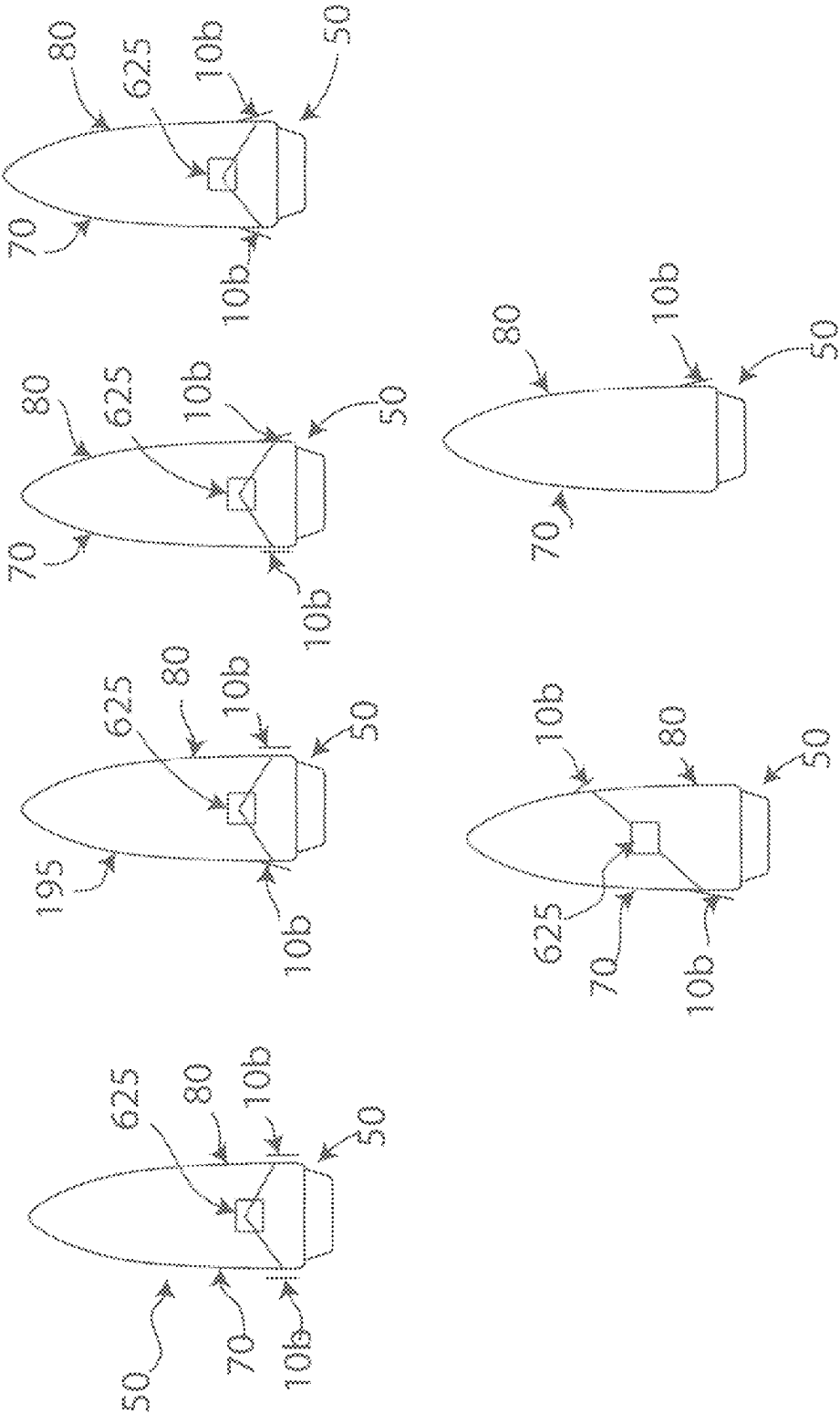


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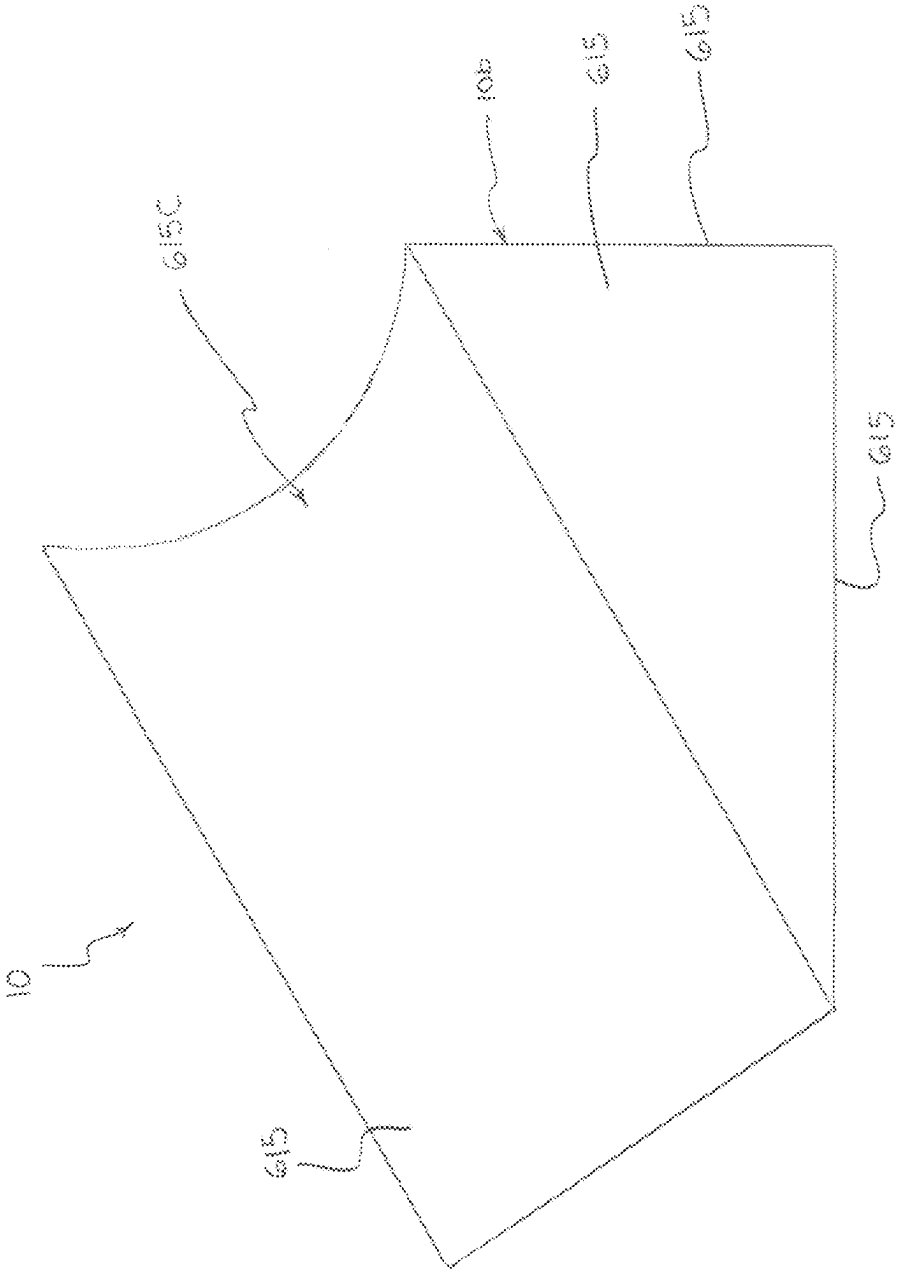


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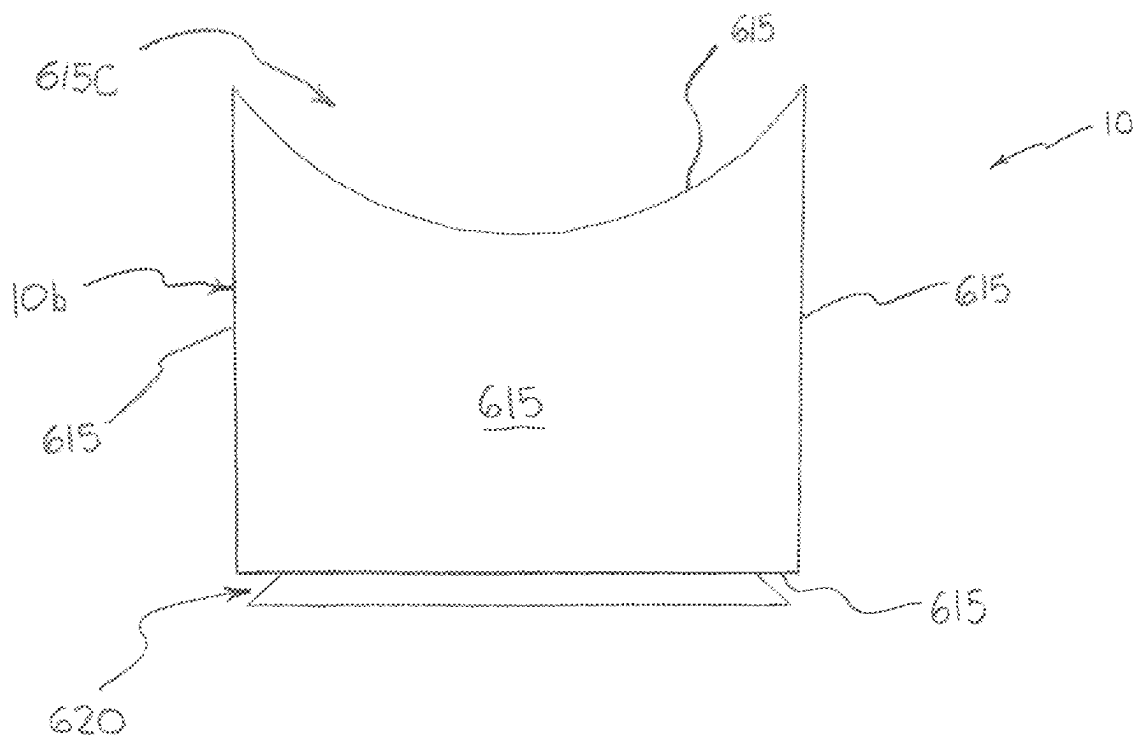
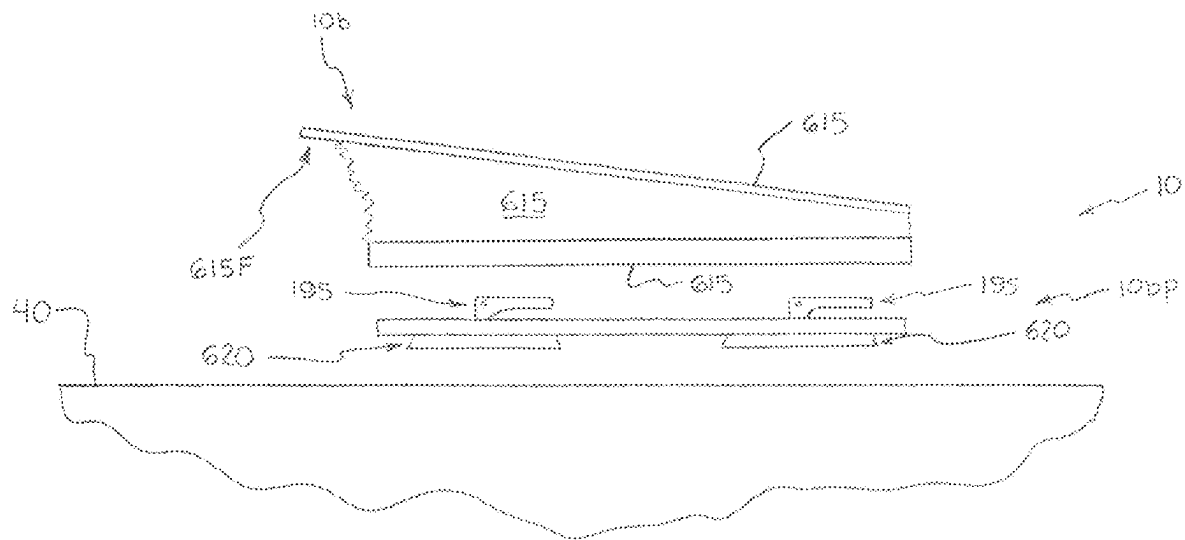


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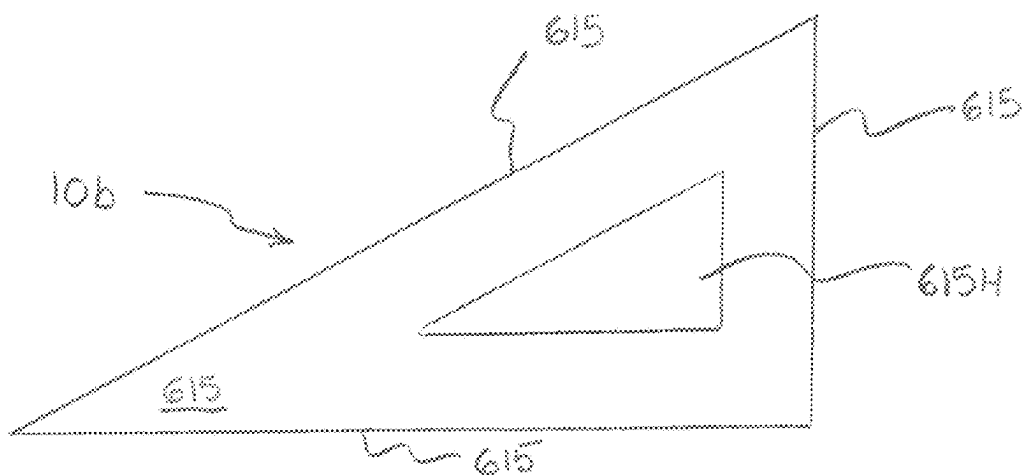


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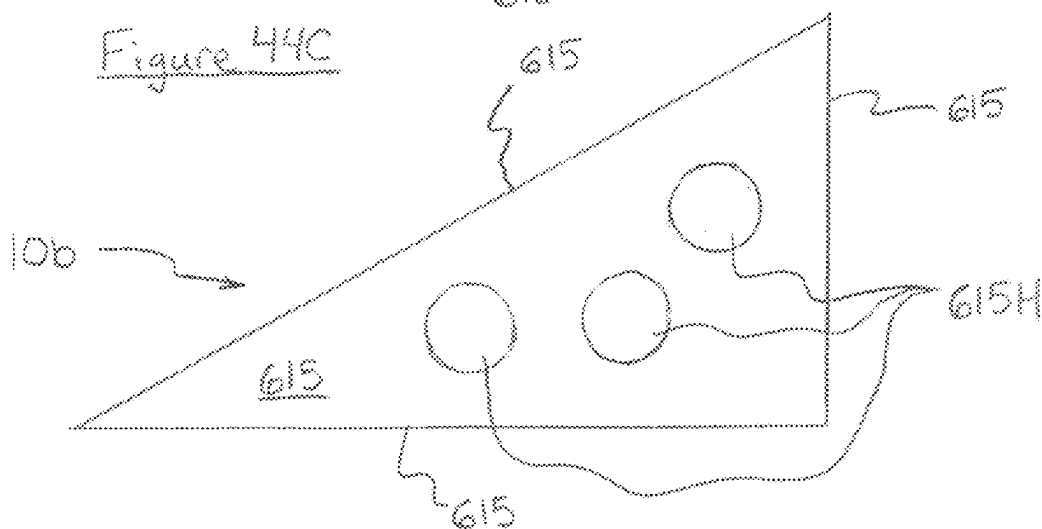


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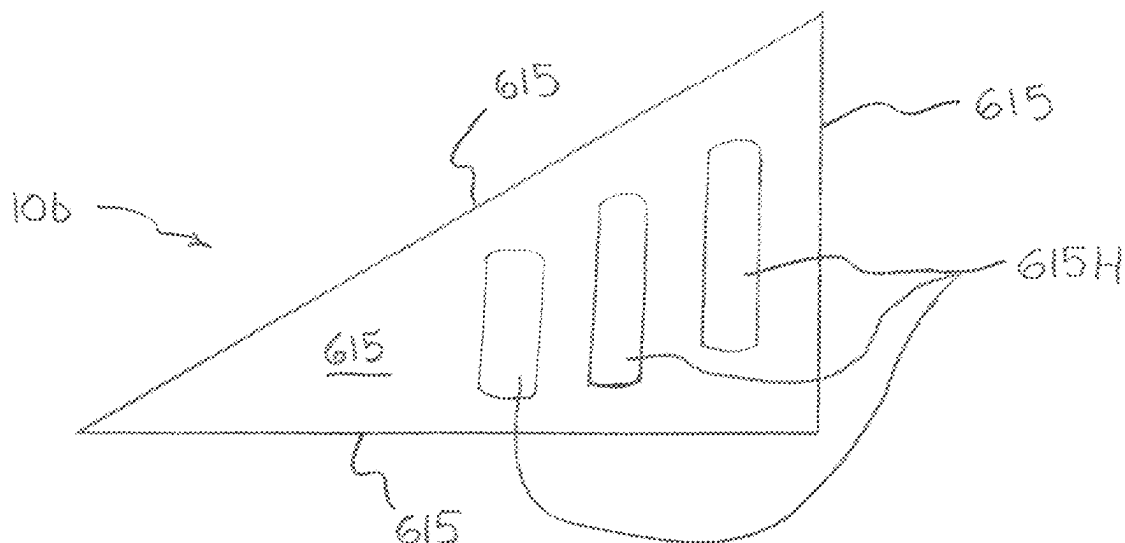
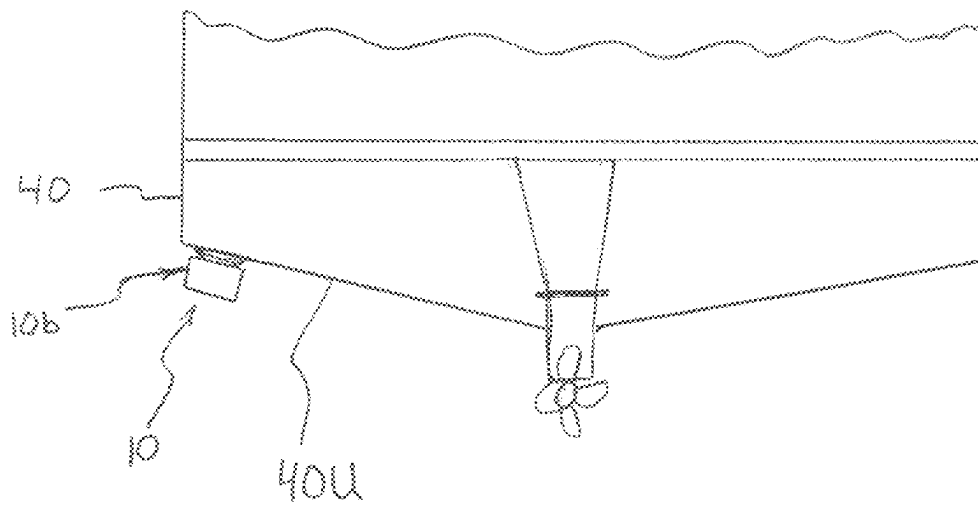


Figure 44A

Figure 45



1

WAKE SHAPING APPARATUS AND RELATED TECHNOLOGY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 16/251,742, filed on Jan. 18, 2019, which is a continuation of U.S. patent application Ser. No. 15/858,779, filed on Dec. 29, 2017, and issued as U.S. Pat. No. 10,183,726, which claims priority to U.S. Provisional Patent Application No. 62/551,621, filed on Aug. 29, 2017, the entire contents of each of which are incorporated by reference herein.

FIELD OF THE INVENTION

The present invention relates to a wake shaping apparatus. More specifically, the present invention provides a wake shaping apparatus configured to attach removably to a hull of a boat such that a wake created behind the boat is larger, differently positioned and/or differently shaped than when the boat is operated without the apparatus. The present invention also provides a deployable lanyard assembly that can be used with the wake shaping apparatus. Further, the invention provides a splash deflector.

BACKGROUND OF THE INVENTION

When a boat moves through water, a wave pattern referred to as a wake is produced on the water surface. In some water sports, such as wake surfing, it is beneficial to modify the size, shape and/or position of a wake created behind the boat in order to enhance the wake surfing experience.

As set forth in the present disclosure, it would be desirable to provide a wake shaping apparatus that removably attaches to a hull of a boat. It would also be desirable to provide a wake shaping apparatus that overcomes the disadvantages of conventional wake shapers known in the prior art. The present invention overcomes such disadvantages, for example, by providing an apparatus that removably attaches to the hull of a boat, is easy to operate, is easily graspable, provides adjustability, and/or provides other advantages. It would also be desirable to provide a splash deflector that can be used with the wake shaping apparatus or on its own. Further, it would be desirable to provide a lanyard assembly that can be connected between the wake shaping apparatus and a boat.

SUMMARY OF THE INVENTION

Certain embodiments of the invention provide an apparatus configured to attach removably to a hull of a boat. The apparatus comprises a base and a blade and has first and second configurations. The first configuration is characterized by the blade being mounted to the base. The second configuration is characterized by the blade being separated from the base. In the present embodiments, the apparatus is configured to provide tool-free adjustment between the first and second configurations.

In some embodiments, the invention provides an apparatus having a handle configuration characterized by an aperture large enough to receive four fingers of a person's hand so as to facilitate manual handling of the apparatus. The apparatus is configured to attach removably to a hull of a boat.

2

Some embodiments of the invention provide an apparatus having generally opposed first and second side regions and a lanyard eyelet. The lanyard eyelet is movable between at least first and second positions. When in the first position, the lanyard eyelet is located at the first side region of the apparatus. When in the second position, the lanyard eyelet is located at the second side region of the apparatus. The apparatus is configured to attach removably to a hull of a boat.

Certain embodiments of the invention provide an apparatus having first and second configurations. The first configuration is characterized by the blade being mounted to the base. The second configuration is characterized by the blade being separated from the base. The blade has opposed first and second faces. When the apparatus is in the first configuration, the blade can be in a first orientation or a second orientation. In the first orientation, the first face of the blade faces a desired direction relative to the base. In the second orientation, the second face of the blade faces the desired direction relative to the base. The blade is equipped with a mount structure that projects outwardly from both of the first and second faces of the blade. The mount structure comprises a first detent that projects from the first face of the blade, and a second detent that projects from the second face of the blade. The first detent is configured to attach to the base of the apparatus such that the blade is retained in the first orientation. The second detent is configured to attach to the base of the apparatus such that the blade is retained in the second orientation. The apparatus is configured to attach removably to a hull of a boat.

Further, some embodiments of the invention provide a combination of a boat and a splash deflector. The boat has a hull to which the splash deflector is mounted removably and noninvasively.

Still further, certain embodiments provide an apparatus comprising an adjustable base. The adjustable base has first and second end regions. The first end region has a first suction cup, and the second end region has a second suction cup. The first end region has a pivot connection with the second end region. The apparatus is configured to attach removably to a hull of a boat.

In other embodiments, the invention provides an apparatus comprising a base and a blade. The blade projects away from the base and has a plurality of channels. The apparatus is configured to attach removably to a hull of a boat.

In still other embodiments, the invention provides an apparatus comprising a base and a blade. The blade projects away from the base and has at least first and second orientations. When in the first orientation, the blade projects away from the base at a first angle. When in the second orientation, the blade projects away from the base at a second angle. The first and second angles are different. The apparatus is configured to attach removably to a hull of a boat.

In certain other embodiments, the invention provides an apparatus configured to attach removably to a hull of a boat such that when attached to the hull, the apparatus is configured to disrupt water flow around the hull of the boat such that a wake created behind the boat during operation of the boat is larger, differently positioned, and/or differently shaped than when the boat is operated without the apparatus attached to the hull. The apparatus comprises a base, a blade and a lanyard assembly. The lanyard assembly comprises a tether portion and a float portion. The float portion comprises a float and a float line. The float line is configured to

3

be attached to the base of the apparatus. The tether portion comprises a tether line configured to be attached to a cleat on the boat.

In still other embodiments, the invention provides a combination of a boat and a wake shaping apparatus. The boat has a hull to which the wake shaping apparatus is mounted. The wake shaping apparatus is inflatable such that the wake shaking apparatus is configured to transition between an inflated state and a non-inflated state.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a broken-away perspective view of a wake shaping apparatus in accordance with certain embodiments of the present disclosure in which the apparatus is attached to a hull of a boat such that a front end of a base of the apparatus faces rearwardly of the boat.

FIG. 2 is a broken-away perspective view of a wake shaping apparatus of the present disclosure similar to FIG. 1, except that the apparatus is attached to the hull of the boat such that a rear end of the base of the apparatus faces rearwardly of the boat.

FIG. 3 is a side view of an embodiment of a wake shaping apparatus of the present disclosure attached to a hull of a boat such that a front end of a base of the apparatus faces rearwardly of the boat.

FIG. 4 is a side view of a wake shaping apparatus of the present disclosure similar to FIG. 3, except that a rear end of a base of the apparatus faces rearwardly of the boat.

FIG. 5 is a side view of an embodiment of a wake shaping apparatus of the present disclosure attached to a hull of a boat at a position that is nearer to a bow than to a stern of the boat.

FIG. 6 is a side view of an embodiment of the present disclosure having two wake shaping apparatuses both attached to the same side of a hull of a boat such that one of the apparatuses is positioned near a bow of the boat and the other of the apparatuses is positioned near a stern of the boat.

FIG. 7 is a front side perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the blade is mounted to the base.

FIG. 8 is a top view of an embodiment of a wake shaping apparatus of the present disclosure.

FIG. 9 is a front side perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the blade is separated from the base.

FIG. 10 is a front view of an embodiment of a wake shaping apparatus of the present disclosure in which a body is in its second position.

FIG. 11 is a front view of an embodiment of a wake shaping apparatus of the present disclosure in which the body is in its first position.

FIG. 12 is a front perspective view of an embodiment of the blade of the present disclosure showing a convex face of the blade.

FIG. 13 is a rear perspective view of an embodiment of the blade of the present disclosure showing a concave face of the blade.

FIG. 14 is a rear view of an embodiment of a wake shaping apparatus of the present disclosure in which a lanyard eyelet is in the first position.

FIG. 15 is a rear view of an embodiment of a wake shaping apparatus of the present disclosure in which the lanyard eyelet is in the second position.

FIG. 16 is a bottom view of an embodiment of a wake shaping apparatus of the present disclosure showing two levers in a locked configuration.

4

FIG. 17 is a partially exploded view of an embodiment of a wake shaping apparatus of the present disclosure showing individual components of two suction cup assemblies.

FIG. 18 is a side view of an embodiment of a wake shaping apparatus of the present disclosure in which each of two levers is in an unlocked configuration.

FIG. 19 is a top view of an embodiment of a wake shaping apparatus of the present disclosure.

FIG. 20 is a top view of an embodiment of the present disclosure having an adjustable base that includes a pivot joint.

FIG. 21 is side view of an embodiment of the present disclosure having an adjustable base that includes another pivot joint.

FIG. 22 is a top view of an embodiment of the present disclosure having an adjustable base that includes a ball joint.

FIG. 23 is a perspective view of a wake shaping apparatus and an embodiment of a splash deflector having an arcuate splash wall, both mounted removably to a hull of a boat.

FIG. 24 is a front view of the splash deflector of FIG. 23.

FIG. 25 is a bottom view of an embodiment of a splash deflector of the present disclosure.

FIG. 26 is a perspective view of a wake shaping apparatus and an embodiment of a splash deflector having a planar splash wall, both mounted removably to a hull of a boat.

FIG. 27 is a perspective view of an embodiment of a splash deflector of the present disclosure having an arcuate splash wall mounted to a transom of a boat.

FIG. 28 is a side view of a blade, a partially exploded side view of an adapter component of the present disclosure, and a perspective view of a base showing a channel configured to selectively receive the first or second detent of the adapter component in certain embodiments of the present disclosure.

FIG. 29 is a side view of an embodiment of the adapter component of the present disclosure showing the adapter component in each of first, second, and third positions.

FIG. 30 is a perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which a blade has an adjustable hinge.

FIG. 31 is a perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which a blade has a virtual hinge, as well as a side view showing the blade attached to the base.

FIG. 32 is a top perspective view of an embodiment of a wake shaping apparatus of the present disclosure having a blade with channels.

FIG. 33 is a detailed perspective view of the embodiment of FIG. 32.

FIG. 34 is a perspective view of an embodiment of both a wake shaping apparatus and a deployable lanyard assembly in which a float portion of the lanyard assembly is attached to the wake shaping apparatus and a tether portion of the lanyard assembly attached to a cleat on a boat.

FIG. 35 is a perspective view of the lanyard assembly of FIG. 34 where the float portion and the tether portion are separated.

FIG. 36 is a side detail view of the deployable lanyard assembly of FIG. 34.

FIG. 37 is a partially exploded side view of the deployable lanyard assembly of FIG. 36.

FIG. 38 shows two views of an embodiment of an inflatable wake shaping apparatus in inflated and non-inflated states so as to schematically illustrate how the apparatus transitions between the two states.

FIG. 39 shows multiple views of another embodiment of an inflatable wake shaping apparatus in both inflated and

5

non-inflated states so as to schematically illustrate how the apparatus transitions between the two states.

FIG. 40 shows multiple views of one or more inflatable wake shaping apparatuses attached to different regions of a boat.

FIG. 41 is a schematic perspective view of an embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

FIG. 42 is a schematic rear view of the wake shaping apparatus of FIG. 41.

FIG. 43 is a schematic side view of another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

FIG. 44A is a schematic side view of still another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

FIG. 44B is a schematic side view of yet another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

FIG. 44C is a schematic side view of still another embodiment of a wake shaping apparatus of the present disclosure in which the wake shaping apparatus is an inflatable apparatus.

FIG. 45 is a broken-way rear view of a wake shaping apparatus attached to an underside of a hull of a boat in accordance with certain embodiments of the present disclosure.

DESCRIPTION OF VARIOUS EMBODIMENTS

The following detailed description is to be read with reference to the drawings, in which like elements in different drawings have like reference numerals. The drawings, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the invention. Skilled artisans will recognize that the examples provided herein have many useful alternatives that fall within the scope of the invention.

Referring to the drawings, and in particular, FIG. 1, there is shown an apparatus of the present disclosure generally represented by reference numeral 10. The illustrated apparatus 10 is a wake shaping apparatus, which preferably is configured to attach removably to a hull 40 of a boat 50. The wake shaping apparatus 10 is configured to disrupt water flow around the hull 40 such that a wake created behind the boat 50 during operation of the boat 50 is larger, differently positioned, and/or differently shaped than when the boat 50 is operated without the apparatus 10 attached to the hull 40.

As shown in FIG. 7, the apparatus 10 generally comprises a base 100 and a blade 300. The apparatus 10 has a first configuration 20 in which the blade 300 is mounted to the base 100, and a second configuration 30 in which the blade 300 is separated from the base 100. As described in greater detail below, in some embodiments, the apparatus 10 is configured to provide tool-free adjustment between the first 20 and second 30 configurations.

The base 100 generally has first 115 and second 120 end regions, and first 125 and second 130 side regions. The illustrated first 125 and second 130 side regions extend between the first 115 and second 120 end regions. The first end region 115 includes a front end 135 (see FIG. 3) of the base 100, and the second end region 120 includes a rear end 140 of the base 100.

6

The base 100 also has a top side 145 and a bottom side 148 (see FIG. 16). The top side 145 is opposite the bottom side 148. Preferably, the bottom side 148 is on the same side of the base 100 as one or more suction cups 105. When the apparatus 10 is attached to the hull 40 of a boat 50, the bottom side 148 faces (and is carried alongside) the hull 40, and the top side 145 faces away from the hull 40.

Referring to FIG. 8, the blade 300 has opposed first 305 and second 310 faces. In some embodiments, the first 305 and second faces 310 are identical, whereas in other embodiments, they are different. For example, the first face 305 of the blade 300 can have different structural features provided thereon and/or a different shape than the second face 310 of the blade 300. Such differences can optionally be provided to create different effects on the wake. In certain embodiments, one of the faces 305, 310 is convex (FIG. 12) and the other of the faces 305, 310 is concave (FIG. 13). However, in other embodiments, both faces 305, 310 of the blade 300 are planar. Alternatively, one face 305, 310 can be planar while the other face 305, 310 is either concave or convex.

As discussed above, the illustrated apparatus 10 has a first configuration 20 (see FIG. 7) in which the blade 300 is mounted to the base 100. When the apparatus 10 is in the first configuration 20, the blade 300 is in either a first orientation or a second orientation. The first face 305 of the blade 300 faces a desired direction relative to the base 300 when the apparatus 10 is in the first orientation. In contrast, the second face 310 of the blade 300 faces the desired direction relative to the base when the apparatus 10 is in the second orientation. It may be advantageous to change the blade 300 between the first orientation and the second orientation when the first 305 and second 310 faces of the blade 300 have different structures and/or shapes in order to achieve different effects on the wake.

The illustrated blade 300 has a mount structure 340 configured to attach to the base 100. The mount structure 340 preferably projects outwardly from the blade 300 (e.g., outwardly from a primary wall portion of the blade). For example, the mount structure 340 can optionally project outwardly from at least one, and preferably both, of the first 305 and second 310 faces of the blade 300. In some alternate embodiments, the base 100 is integral to (or permanently affixed to) the blade 300. However, in preferred embodiments, the base 100 and the blade 300 are configured to be attached to each other removably. Where the mount structure 340 projects outwardly from both faces 305, 310 of the blade 300, the blade 300 is configured to be removably attached to the base 100 in either of two orientations, i.e., in either the first or second orientation so that a desired one of the faces 305, 310 of the blade 300 faces the desired direction.

Preferably, the apparatus 10 is configured to be adhered to the hull 40 of the boat 50 by at least one suction cup 105 so as to removably attach the apparatus 10 to the hull 40 of the boat 50. However, the apparatus 10 can alternatively be removably attached to the hull 40 through other non-invasive forms of mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like. The removable nature of the apparatus 10 allows it to be used on a boat 50 that already has a built-in wake-shaping device so as to complement the effects provided by the built-in wake-shaping device. Alternatively, the apparatus 10 can be used on a boat 50 that does not have any other wake-shaping devices.

Thus, the apparatus 10 preferably can be attached removably to the boat 50 in a desired orientation and on a desired region of the boat 50. Both the orientation of the apparatus 10 when attached to the boat, and the region of the boat to which the apparatus 10 is attached, can be selected to

achieve particular effects on the wake. For example, in FIGS. 1 and 3, the apparatus 10 is attached to the hull 40 of the boat 50 such that the front end 135 of the base 100 faces rearwardly of the boat 50. FIGS. 2 and 4 show the reverse orientation, i.e., the rear end 140 of the base 100 faces rearwardly of the boat 50. The apparatus 10 can be attached to the hull 40 of the boat 50 at a position that is adjacent to a stern 90 (FIGS. 1-4) or adjacent to a bow 95 (FIG. 5) of the boat 50. Additionally or alternatively, a wake shaping apparatus 10 can be attached to a bottom (or underside) 40U of the hull 40. Reference is made to FIG. 45. Any embodiment of a wake shaping apparatus 10 described in this disclosure can optionally be attached releasably to the bottom 40U of the hull 40. This can optionally be in addition to there being one or more wake shaping apparatuses provided on other locations of the hull. Alternatively, the only wake shaping apparatus 10 attached releasably to the hull may be on the bottom (or underside) 40U of the hull 40. This can optionally be the case in FIG. 45. While FIG. 45 shows the wake shaping apparatus 10 as being an inflatable apparatus 10b, this is by no means required. To the contrary, FIG. 45 represents a group of embodiments wherein any wake shaping apparatus is attached releasably to the bottom (or underside) of a hull of a boat. In FIG. 45, the illustrated wake shaping apparatus 10 is attached to the bottom of the hull by a plurality of suction cups. However, various other releasable attachment means can be used instead.

In some embodiments, two wake shaping apparatuses 10 are each attached to the same side of a hull 40 of a boat 50. As shown in FIG. 6, one of the apparatuses 10 can be positioned adjacent to the bow 95 of the boat 50, while the other apparatus 10 is positioned adjacent to the stern 90 of the boat 50. Given the present teaching as a guide, skilled artisans will appreciate that the one or more apparatuses 10 can be mounted removably at various locations on the hull. Tool-Free Adjustment

In certain embodiments, the base 100 and the blade 300 can be separated from each other to achieve the second configuration 30 described above. One example of the second configuration 30 is shown in FIG. 9. Preferably, when the apparatus 10 is in the second configuration 30, the base 100 and the blade 300 can be nested alongside each other so as to occupy a more compact (i.e., smaller) space than when the apparatus 10 is in the first configuration 40. In certain embodiments, the apparatus 10 when in the second configuration can have its blade and base nested alongside each other so as to fit within a rectangular volume (e.g., inside a box) having the following dimensions: 15 inches by 6 inches by 9 inches, or even 14 inches by 5 inches by 8 inches. While not in use, the apparatus 20 can be removed from the hull 40 of the boat 50 and stored conveniently until needed.

In some embodiments, the apparatus 10 can be adjusted from the first configuration 20 to the second configuration 30, and from the second configuration 30 to the first configuration 20, without using any tools, and preferably without removing or inserting any fastener from or into the base 100. In such cases, the blade 300 can be mounted to the base 100 and later removed therefrom without using any tools. This is particularly advantageous since the apparatus 10 is intended to be attached to the hull of a boat 50 at a location that is under water while the boat 50 is in operation. Using tools underwater can prove difficult due to visibility issues, the tendency of tools to become slippery when wet, and/or the potential for the water to be cold. These factors may lead to tools and/or fasteners being dropped into the water.

In some embodiments, an optional hook and loop fastener is sandwiched between the blade 300 and the base 100 when the apparatus 10 is in the first configuration 20. In such embodiments, the hook and loop fastener is configured to removably fasten the blade 300 to the base 100. When provided, the hook and loop fastener preferably is used in addition to the mount structure 340. However, in some embodiments, the hook and loop fastener may replace the mount structure 340 entirely.

When provided, the hook and loop fastener includes a first portion attached to the base 100 and a second portion attached to either the mount structure 340 or directly to one or both faces 305, 310 of the blade 300. Preferably, the second portion of the hook and loop fastener is attached to the mount structure 340 on both the first 305 and second 310 faces of the blade 300 so that the blade 300 can be positioned in either the first 315 or second 320 orientation as desired.

As discussed above, it is preferable that the apparatus 10 is configured to be removably attached to a hull of a boat using one or more suction cups 105. In such cases, at least one suction cup 105 on the base 100 forms part of a suction cup assembly 110. Preferably, the suction cup assembly 110 includes at least two (or has exactly two) suction cups 105. However, the suction cup assembly 110 can alternatively include only one suction cup 105, or it can include more than two suction cups 105.

In preferred embodiments, the base 100 further includes a detent 150 having at least first 155 and second 160 positions. In the first position 155, the detent 150 is configured to lock the blade 300 on the base 100. In the second position 160, the detent 150 is configured to release the blade 300 from the base 100. One example of the second position 160 is shown in FIG. 10, and one example of the first position 155 is shown in FIG. 11. Preferably, the detent 150 is rotatable between its first 155 and second 160 positions. However, it should be appreciated that in some embodiments, the detent 150 is not limited to rotational movement, and can instead move linearly, for example. In some embodiments, the detent 150 is part of a suction cup assembly 110 and is located between the base 100 and a suction cup 105.

The illustrated base 100 (e.g., a shoulder thereof) has a channel 165 therein. In a preferred embodiment, the channel 165 is a slot. The illustrated blade 300, and more specifically an adapter component 325 of the illustrated blade, comprises at least one plate, and preferably two plates 355, 365, each configured to be received removably in the channel 165. The adapter component 325 and the channel 165 are shown in FIG. 28. Preferably, the two plates of the adapter component 325 include a first plate 355 disposed adjacent the first face 305 of the blade 300, and a second plate 365 disposed adjacent the second face 310 of the blade 300. The adapter component 325, however, can alternatively have only a single plate, which can be located on either the first face 305 or second face 310 of the blade 300.

The base 100, and more specifically the detent 150, can comprise a body 170 that is moveable between first 175 and second 180 positions. When in the first position 175 (see FIG. 11), the body 170 at least partially closes the channel 165 of the base 100. When in the second position 180 (see FIG. 10), the body 170 is spaced from the channel 165 of the base 100. In a preferred embodiment, the body 170 comprises a rotatable flange 185 that is rotatable between its first 175 and second 180 positions. However, it should be appreciated that in some embodiments, the body 170 is not limited to rotational movement, and can instead move linearly, for example. The body 170 can optionally comprise a

generally circular disc **190** from which the rotatable flange **185** projects. In preferred embodiments, the body **170** is part of a suction cup assembly **110** and the disc **190** is located between a suction cup **105** and the base **100**.

When provided, the suction cup assembly **110** can optionally include at least one lever **195** moveable between first **200** and second **205** positions. Preferably, one lever **195** is coupled to each respective suction cup **105** so as to be operable to activate and release suction for the respective suction cup **105**. However, it is possible that a single lever could be provided that simultaneously controls the suction force for all or a select group of multiple suction cups. Also, as noted above, there may be only a single suction cup on the apparatus in some cases. When the one or more levers **195** are each in the first position **200** (see, e.g., FIG. **8**), each suction cup **105** is in a locked configuration. The locked configuration helps secure the respective suction cup **105** to the hull **40** of the boat **50**. When the at least one lever **195** is in the second position **205** (see FIG. **18**), the respective suction cup **105** is in an unlocked configuration. The unlocked configuration allows the respective suction cup **105** to be released from the hull **40** of the boat **50**. Two levers **195** are each shown in a locked configuration in FIGS. **7**, **8** and **19**, and in an unlocked configuration in FIG. **18**.

In some embodiments, two suction cups **105** are disposed in a confronting angle configuration. In such cases, the confronting angle configuration is characterized by a tilt angle (e.g., relative to horizontal and/or desired plane) of greater than zero degrees and up to three degrees. By having two suction cups **105** in such a confronting angle configuration, the suction cups **105** may adhere particularly stably to a curved surface of a hull **40** of a boat **50**.

It is to be appreciated that tool-free adjustment need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with tool-free adjustment of the nature described above.

Handle Configuration

In some embodiments, the apparatus **10** has a handle configuration **210** characterized by an aperture **215** in the apparatus **10**. Preferably, the aperture **215** is in the base **100**, as shown in FIG. **16**. The aperture **215** is sized to facilitate manual handling and grasping of the apparatus **10**. In a preferred embodiment, the aperture **215** is large enough (e.g., is configured) to receive four fingers of a person's hand. Preferably, the aperture **215** is elongated and/or has a generally rectangular or generally oval shape. However, it is contemplated that the aperture **215** can have any shape as long as it does not interfere with the aperture's intended purpose. In certain embodiments, the aperture has a length of at least two inches, or at least three inches, such as between three inches and six inches.

In a preferred embodiment, the aperture **215** has an aspect ratio of greater than two. As used in the present disclosure, the aspect ratio is defined as a length of the aperture **215** divided by a width of the aperture **215**. The length of the aperture **215** is measured along an axis extending between the front end **135** and the rear end **140** of the base **100**. The width of the aperture **215** is measured along an axis extending between the first **115** and second **120** side regions of the base **100**.

The base **100** preferably has two ribs **230** that extend between (optionally entirely between) the front end **135** and rear end **140** of the base **100**. In preferred embodiments, the aperture **215** is located (e.g., defined) between the ribs **230**. Preferably, both of the ribs **230** are defined by a single integral body. However, in other embodiments, the ribs **230**

can be separate and distinct structures from each other. Each rib **230** preferably is elongated, e.g., so as to have a length longer than six inches, such as between six and 14 inches.

As discussed above, the base **100** has a top side **145** and a bottom side **148**. Preferably, the aperture **215** passes entirely through the base **100** from its bottom side **148** to its top side **145**. However, in other embodiments, the aperture **215** is a blind opening such that the aperture **215** passes only partially between the bottom side **148** and the top side **145** of the base **100** (e.g., so as to have a closed bottom).

As also discussed above, in some embodiments, the suction cup assembly **110** includes two levers **195**, each of which is associated with a respective suction cup **105**. Such levers **195** can advantageously be disposed in a recessed area **220** (see FIG. **7**) of the base **100**. The recessed area **220** can, for example, define a channel **225** that extends between two ribs **230** of the base **100** and is open to aperture **215**. Both of the illustrated levers **195** are aligned with the aperture **215** when in their first (locked) positions **200**. Preferably, the levers **195** do not hinder (or at least do not prevent) manual access to the aperture **215**. Additionally, when in the first position **200**, the levers **195** can optionally be located entirely within the recessed area **220**, e.g., so that the levers **195** are sheltered from being inadvertently moved to the second position **205** and thus unlocked.

It is to be appreciated that the handle configuration need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be so configured.

Lanyard Eyelet

The apparatus **10** preferably has a lanyard eyelet **235**. In such cases, in order to tie the apparatus **10** to a boat **50**, a tether **500** can be removably attached to both the boat **50** and the lanyard eyelet **235**. The tether **500** can comprise a rope, bungee cord or the like.

When provided, the lanyard eyelet **235** preferably is movable between at least first **240** and second **245** positions. When in the first position **240**, the lanyard eyelet **235** is located at the first side region **125** of the apparatus **10**. When in the second position **245**, the lanyard eyelet **235** is located at the second side region **130** of the apparatus **10**. FIG. **14** shows the lanyard eyelet **235** in the first position **240**, and FIG. **15** shows the lanyard eyelet **235** in the second position **245**. Providing for such movement of the lanyard eyelet **235** is advantageous since it can be positioned as needed for convenient attachment thereto by a tether **500**. For example, when the front end **135** of the base **100** faces rearwardly of the boat **50**, the lanyard eyelet **235** can be moved to the first position **240**, and thus, to the first side region **125** of the base **100**. On the other hand, when the rear end **140** of the base **100** faces rearwardly of the boat **50**, the lanyard eyelet **235** can be moved to the second position **245**, and thus, to the second side region **130** of the base **100**. This may make it possible to use a shorter length of tether, which may be advantageous. It can also help keep the tether up out of the water flow path along the wake shaping apparatus.

Preferably, the apparatus **10** includes a suction cup assembly **110** comprising a disc **250** that is rotatable between first **240** and second **245** positions (see FIGS. **14** and **15**). In preferred embodiments of this nature, the lanyard eyelet **235** can optionally be part of the suction cup assembly **110** and may be located between a suction cup **105** and the base **100**. In such cases, the lanyard eyelet **235** preferably is a projection of the rotatable disc **250**.

In another embodiment, the suction cup assembly **10** does not include rotatable disc **250**. In this embodiment, the

11

lanyard eyelet **235** can be fixedly or removably attached to any portion of the apparatus **10**.

In embodiments where the apparatus **10** includes a suction cup assembly **110** having two suction cups **105**, the rotatable disc **250** and a generally circular disc **190** can each be positioned in a stacked orientation and located between a respective suction cup **105** and the base **100**. Alternatively, in another embodiment having only a single suction cup **105**, rotatable disc **250** can be omitted, while a generally circular disc **190** is positioned between the single suction cup **105** and the base **100**.

A detailed view of one example of a suction cup assembly **110** is shown in FIG. 17. Here, each lever **195** of the suction cup assembly **110** has a hole **505** disposed therein. A pin **510** is provided for each lever **195** such that one of the pins **510** can be received in the hole **505** of the respective lever **195**. Each suction cup **105** further has a post **515** attached thereto and extending outwardly therefrom. Each post **515** has a hole **520** disposed therein. In addition to passing through the hole **505** of the respective lever **195**, each pin **510** also passes through the hole **520** of the post **515** of the respective suction cup **105**. Additional holes (not shown) are provided in the base **100**, disc **190**, and disc **250** to allow the levers **195** to attach to the respective suction cups **105**. The suction cup assembly **110** also includes springs **525**, optionally such that one spring **525** is provided for each lever **195**. The springs **525** are located between the levers **195** and their respective suction cups **105**.

It is to be appreciated that the lanyard eyelet need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with an adjustable lanyard eyelet of the nature described above.

Mount Structure

As discussed above, the apparatus **10** preferably has different configurations depending on whether the base **100** is mounted to the blade **300** (first configuration **20**) or separated from the blade **300** (second configuration **30**). Additionally, when the apparatus **10** is in the first configuration **20**, the blade **300** preferably is in either a first orientation or a second orientation, depending on which face **305**, **310** of the blade **300** faces the desired direction relative to the base **100**.

As also discussed above, the blade **300** can advantageously be equipped with a mount structure **340** that projects outwardly from at least one, and preferably both, faces **305**, **310** of the blade **300**. In some embodiments, the mount structure **340** comprises a first detent **345** that projects from the first face **305** of the blade **300**, and a second detent **350** that projects from the second face **310** of the blade **300**. The first detent **345** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the first orientation **315**. The second detent **350** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the second orientation **320**.

In certain embodiments, the first **345** and second **350** detents are fixed in place on (i.e., so as not to be movable relative to) the blade **300**. In other embodiments, however, the first **345** and second **350** detents are part of an adapter component **325** that is moveable (e.g., relative to a primary wall of the blade **300**) between at least first **330** second **335** positions. In the first position **330**, the adapter component **325** is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the first orientation **315**. In the second position **335**, the adapter component **325**

12

is configured to attach to the base **100** of the apparatus **10** such that the blade **300** is retained in the second orientation **320**.

In some embodiments, the blade **300** is configured to lie at least generally in a plane (e.g., has a configuration that is at least generally planar), and the adapter component **325** is configured to move at least generally perpendicular to such plane when the adapter component **325** moves between its first **330** and second **335** positions. The adapter component **325** can also move to an intermediate position **338** that is in-between the first **330** and second **335** positions. The first **345** and second **350** detents can optionally be connected together by conventional fasteners, such as nuts and bolts. Such fasteners can allow the first **345** and second **350** detents to slide between the first **330** and second **335** positions.

As shown in FIG. 29, when the adapter component **325** is in the first position **330**, the first detent **345** projects further from the first face **305** of the blade **300** than it does when the adapter component **325** is in the second position **335**. Likewise, when the adapter component **325** is in the second position **335**, the second detent **350** projects further from the second face **310** of the blade **300** than it does when the adapter component **325** is in the first position **330**. FIG. 29 shows the adapter component **325** in the first position **330** (image on the left), the second position **335** (middle image), and the intermediate position **338** (image on the right).

The first detent **345** of the adapter component **325** can optionally comprise a first plate **355** having a first mounting flange **360**. Similarly, the second detent **350** of the adapter component **325** can optionally comprise a second plate **365** having a second mounting flange **370**. Preferably, the first **355** and second **365** plates are moveable conjointly relative to the first **305** and second **310** faces of the blade **300**. When the blade **300** is positioned in the first orientation **315**, the first mounting flange **360** is received in the channel **165**. When the blade **300** is positioned in the second orientation **320**, the second mounting flange **370** is received in the channel **165**.

As discussed above, the base **100** has opposed first **115** and second **120** end regions. The first end region **115** (e.g., a shoulder **255** thereof) defines the channel **165**. In a preferred embodiment, the first end region **115** has a greater height than the second end region **120**. Alternatively, in some embodiments, the first and second end regions can be identical in height, or the second end region can have a taller height than the first end region so long as the second end region includes the channel.

It is to be appreciated that the preferred blade mount structure need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with a blade mount structure of the nature described above.

Adjustable Base

In certain embodiments, the base **100** is an adjustable base **100a**. In some embodiments of this nature, the first end region **115** of the adjustable base **100a** has a first suction cup **105a**, and the second end region **120** of the adjustable base **100a** has a second suction cup **105b**. To make such a base adjustable, the first end region **115** can optionally have a pivot connection **260** with the second end region **120**. In some cases, the pivot connection **260** allows the suction cups **105a**, **105b** to be oriented at various angles relative to each other. In this manner, the pivot connection **260** may permit the suction cups **105a**, **105b** to be stably mounted to surfaces of different contours, including both flat and curved surfaces. For example, the suction cups **105a**, **105b** may be configured to be positioned in a confronting angle configura-

13

ration such that the suction cups **105a**, **105b** are angled somewhat inwardly toward each other.

When provided, the pivot connection **260** preferably comprises a first pivot joint having a first pivot axis. In FIG. **20**, the first pivot joint enables adjusting an angular orientation of the second end region **120** relative to that of the first end region **115** such that the apparatus **10** has at least first and second angular configurations. The first angular configuration is characterized by the first and second end regions extending along a straight line (see FIG. **19**). The second angular configuration is characterized by one of the first **115** and second **120** end regions projecting away from the other at a positive acute angle (see FIG. **20**). In this embodiment, both the first **105a** and second **105b** suction cups can remain in a constant plane while the apparatus **10** is moved between its first and second angular configurations.

Referring now to FIG. **21**, the pivot connection **260** can optionally comprise a second pivot joint having a second pivot axis. In other embodiments, the pivot connection **260** includes both the first and second pivot joints. In FIG. **21**, the pivot connection has only the second pivot joint. The second pivot joint enables pivoting the apparatus **10** in either a first direction or a second direction. In particular, when the apparatus **10** is pivoted in the first direction, the first **105a** and second **105b** suction cups pivot toward each other. When the apparatus **10** is pivoted in the second direction, the first **105a** and second **105b** suction cups pivot away from each other.

The pivot connection **260** can optionally comprise a ball joint, as shown in FIG. **22**. When provided, the ball joint preferably is located between the first **115** and second **120** end regions of the adjustable base **100a** and preferably is configured to enable pivotal movement between the first **115** and second **120** end regions.

It is to be appreciated that the adjustable base structure need not be provided in all embodiments of the invention. However, any embodiment of the present disclosure can optionally be provided with an adjustable base structure of the nature described above.

Splash Deflector

The invention also provides embodiments wherein a splash deflector **265** is mounted removably and non-invasively to a boat **50**. The splash deflector **265** is intended to limit the splash of water adjacent to a region of the boat **50** where the splash deflector **265** is attached. The illustrated boat **50** has a transom **60** that extends between opposed sides **70**, **80** of the hull **40**. The splash deflector **265** can be mounted to the transom **60**, as shown in FIG. **27**, or to a desired one of the sides **70**, **80** of the hull **40**, as shown in FIGS. **23** and **26**.

If desired, first and second splash deflectors can be mounted respectively on first and second sides of the hull. Additionally or alternatively, first and second splash deflectors can be mounted on the transom of a boat. Given the present teaching as a guide, skilled artisans will appreciate that one or more splash deflectors can be provided at various locations on one or both sides of the hull and/or at various locations on the transom.

The splash deflector **265** comprises both a splash wall **270** and a mount wall **275**. The splash wall **270** is attached to and projects outwardly from the mount wall **275**. When the splash deflector **265** is attached to the hull **40** of a boat **50**, the mount wall **275** is carried alongside the hull **40**, and the splash wall **270** projects away from the hull **40**. As shown in FIG. **26**, the splash wall **270** can be planar, or at least generally planar, such that it extends away from the mount

14

wall **275** in a generally perpendicular manner. Alternatively, as shown in FIGS. **23** and **27**, the splash wall **270** can be arcuate, e.g., so as to form a curved fender configuration.

The splash deflector **265** is removably mounted to a boat **50**, preferably by one or more suction cups **280**. In a preferred embodiment, the mount wall **275** of the splash deflector **265** includes two suction cups **280**, as shown in FIG. **25**. However, in other embodiments, the splash deflector **280** has only a single suction cup **280**, or it has more than two suction cups **280**. Moreover, instead of suction cups **280**, the splash deflector **265** can be attached to the boat by other non-invasive and removable forms of mechanical attachment, such as hook and loop fasteners, magnets or the like.

Similar to the wake shaping apparatus **10**, the splash deflector **265** can optionally include one or more levers **285** that control suction of the preferred suction cups **280**. Preferably, one lever **285** is provided for each suction cup **280**, as shown in FIG. **24**. However, a single lever could alternatively be provided that controls suction for all suction cups on the mount wall. The lever **285** can be moved between a locked configuration and an unlocked configuration. In the locked configuration, each lever **285** activates suction for the respective suction cup **280**. In the unlocked configuration, each lever **285** releases suction for the respective suction cup **280**.

Preferably, the splash deflector **265** has an aperture **290** passing therethrough. When provided, the aperture **290** preferably is sized to facilitate manual handling and grasping of the splash deflector **265**. In such cases, the aperture **290** is configured to receive four fingers of a person's hand. The mount wall **275** has a first surface **295** and a second surface **298**. Preferably, the aperture **290** passes entirely through the mount wall **275** from its first surface **295** to its second surface **298**, as shown in FIG. **23**. In other embodiments, however, the aperture comprises a blind bore such that the aperture passes only partially between the first surface and the second surface of the mount wall.

The splash deflector **265** can be used with or without the wake shaping apparatus **10**. When the splash deflector **265** and the wake shaping apparatus **10** are used in combination, the splash deflector **265** is positioned above the apparatus **10**, as shown in FIG. **27**. Such positioning of the splash deflector **265** can advantageously limit the amount of water from the apparatus **10** that splashes into the boat **50**.

In certain embodiments, the splash deflector and the wake shaping apparatus are two discrete components that can be removably attached to each other during use. Removable attachment of the splash deflector and the wake shaping apparatus can be accomplished by various conventional means, including, but not limited to, a mounting wall (not shown) attached to both the splash deflector and the apparatus by screws or other fasteners.

In other embodiments, the splash deflector and the apparatus are integrally formed such that they define a one-piece structure. For example, a mounting wall can be provided that fixedly attaches the splash deflector to the wake shaping apparatus.

Adjustable Blade

In some embodiments, an angular orientation of the blade **300** (e.g., relative to the base **100**) can be adjusted. This feature allows the blade **300** to be adjusted so as to project away from the base **100** at different angles to provide various effects on the wake. In the present embodiments, the blade **300** has at least first and second orientations. In the first orientation, the blade **300** projects away from the base **100** at a first angle. In the second orientation, the blade **300**

15

projects away from the base **300** at a second angle. The first and second angles are different. Preferably, the blade **300** is pivotable between the first and second orientations via a hinge **380**. The hinge **380** can be a simple hinge, as shown in FIG. **30**, or a virtual hinge, as shown in FIG. **31**.

The blade **300** further includes a mounting plate **385**. The mounting plate **385** is attached to the blade **300** (e.g., is attached to a primary wall portion of the blade), e.g., via the hinge **380**, preferably such that the hinge **380** enables the blade **300** (or a primary wall portion thereof) to pivot relative to the mounting plate **385**. Similar to the first **355** and second **360** plates described above, the mounting plate **385** can be removably mounted in a channel **165** defined by the base **100**. In some embodiments, the mounting plate **385** is attached directly to the blade **300** (e.g., to a primary wall portion thereof) via the hinge **380**.

As shown in FIG. **30**, an optional base plate **390** can be provided. The base plate **390**, for example, can be attached to the blade **300** (e.g., to a primary wall portion thereof) between the blade **300** (e.g., a primary wall portion thereof) and the mounting plate **385**. In FIG. **30**, the mounting plate **385** is hingedly attached to the base plate **390** such that the mounting plate **385** is pivotable relative to the base plate **390**.

As discussed above, FIG. **31** shows a virtual hinge allowing the blade **300** to pivot between the at least first and second orientations. In some embodiments having a virtual hinge, the mounting plate **385** has a projection **395** extending outwardly therefrom. The projection **395** of the mounting plate **385** can have a slot **398** disposed therein. Preferably, the projection **395** is a semi-circle, and/or the slot **398** is arcuate, as shown in FIG. **31**. However, other shapes for both the projection **395** and the slot **398** are contemplated.

In a preferred embodiment, a knob and bolt structure is provided. In certain embodiments, two walls (not shown) are attached to and extend between the blade **300** and the mounting plate **385**. One or more bolts can be attached to the walls. One or two knobs can be provided, and each knob is attached to an end of the respective bolt. The bolt(s) are received in the slot **398** and are movable along a length of the slot **398**. Tightening of the bolts permits the mounting plate **385** and the blade **300** to be locked at a desired angle relative to each other.

Blade with Channels

Referring to FIGS. **32** and **33**, in certain embodiments the blade **300** has a plurality of channels **400**. The channels **400** preferably are integral to the blade **300** in that the blade **300** defines walls **405** bounding the channels **400**. When the blade **300** is attached to the base **100**, the channels **400** (or at least some of the channels) on the blade **300** preferably extend generally away from the base **100**.

The channels **400** have a configuration designed to impact water flow along the blade. For example, the shape, size and orientation of the channels **400** may affect the shape of the wake. The channels **400** can be provided on either or both faces **305**, **310** of the blade **300**. The channels **400** can have different configurations on each face **305**, **310** in order to provide different effects depending upon which face **305**, **310** of the blade **300** faces the base **100**.

The channels **400** on a single blade **300** can all be uniform in size. Alternatively, the channels **400** on the blade **300** can be different in size. Where the channels **400** on the blade **300** are uniformly sized, the channels **400** can all be customized so they are smaller or larger as desired to affect performance.

Additionally, the walls defining the channels **400** can optionally project outwardly from the blade **300** at a 90-degree angle, or they can form any angle with the blade **300**

16

that is greater than zero. In certain embodiments, all the channels **400** on a given side of the blade are uniform in orientation and size. Alternatively, in some embodiments, the orientations of certain channels **400** on the blade **300** and/or the sizes of certain channels **400** are not uniform. Certain embodiments provide one side of the blade with channels of a given configuration, while channels on the other side of the blade have a different configuration.

In FIGS. **32** and **33**, all of the channels **400** are at least generally parallel to one other. Thus, the channels **400** can optionally be parallel to one another along an entire lengthwise extent of the blade **300**. Although the channels **400** are shown as extending in straight lines across the blade **300**, at least some (or all) of the channels **400** may instead be curved, e.g., one or more of them may have curved portions.

The illustrated blade **300** further includes a mount region **410** configured to be carried against the base **100**. The mount region **410** preferably is devoid of channels **400** so as not to interfere with mounting the blade **300** to the base **100**. Depending on the particular embodiment, the mount region **410** can be an area on the blade **300** where an optional adapter component **325**, mount structure **340**, mounting plate **385**, and/or base plate **390** are disposed.

Deployable Lanyard Assembly

In some embodiments, the invention provides a lanyard assembly **530**. Referring to FIGS. **34-37**, one example of the lanyard assembly **530** comprises a tether portion **535** and a float portion **540**. The tether portion **535** comprises a tether line **555** configured for attaching to a cleat **560** on a boat **50**. The tether line **555** can comprise a rope, bungee cord or the like. The float portion **540** comprises both a float **545** and a float line **550**. The float line **550** can comprise a rope, bungee cord or the like and is configured for attaching to the base **100** of the apparatus **10**. In some cases, the float line **550** can be attached to a lanyard eyelet **235** of the base **100**.

Preferably, the tether portion **535** and the float portion **540** are configured to be connected (and in FIG. **34**, are connected) to each other by a friction joint. For example, the tether portion **535** can optionally comprise a male detent (e.g., a tongue) **565** and the float portion **540** can comprise a female detent **570**. Alternatively, the male detent can be provided on the float portion while the female portion is provided on the tether position. In such cases, the male **565** and female **570** detents collectively define the friction joint. Various other friction joint structures can alternatively be used.

The tether line **555** has first **575** and second **580** ends. The first end **575** of the tether line **555** is configured to be attached to a cleat **560** on a boat **50**. The second end **580** of the tether line **555** can optionally have a tongue **585**. The float line **550** also has first **590** and second **595** ends. The first end **590** of the float line **550** is configured to be attached to the float **545**. The second end **595** of the float line **550** is configured to be attached to the base **100**. For example, in some cases the second end of the float line **550** is configured to be attached to a lanyard eyelet **235** of a wake shaping apparatus **10**. The float **545** has front **600** and rear **605** ends. In the embodiment illustrated, the rear end **605** of the float **545** has an opening **610**. The second end **595** of the illustrated float line **550** also has an opening **615**. The tongue **585** of the tether line **555** can be received in both openings **610**, **615** so as to provide a friction fit therebetween.

The float **545** is buoyant such that it is configured to float on water. In a preferred embodiment, the float **645** is a foam material (e.g., a closed foam). However, the float **645** can comprise virtually any buoyant material, e.g., an inflatable plastic, so long as the float **545** is capable of providing

17

buoyancy in water. As shown in FIGS. 36 and 37, the float 545 has a bore (e.g., an axial bore) 630 extending between the front 600 and rear 605 ends thereof. Here, the tether line 555 extends through the bore 630 of the float 545. Inflatable Wake Shaper Apparatus

In certain embodiments, the invention provides a wake shaping apparatus 10 that is an inflatable apparatus 10b. In such cases, the apparatus (at least when inflated) preferably can float on water. As used in the present disclosure, the term “inflatable apparatus” refers to an apparatus that can transition between a non-inflated state and an inflated state. The means for inflating the inflatable apparatus 10b can be air or other gas, water or other liquid, or a combination of gas and liquid. In some cases, the apparatus 10b is configured to have a varied internal pressure of gas or liquid (i.e., it can be inflated to different extents) to create multiple but intentional effects, such as different shapes and/or angles of the apparatus 10b.

Referring to FIGS. 38 and 39, the inflatable apparatus 10b has a plurality of walls 615. In some embodiments, none of the walls 615 are rigid. In other embodiments, one or more of the walls 615 are rigid to help define a shape of the apparatus 10b. For example, while being inflated, the apparatus 10b can optionally expand until one of the rigid walls forms a desired angle with an adjacent non-rigid wall. The extent of expansion, and thus, the angle the rigid wall forms with the adjacent non-rigid wall, can optionally depend on a volume of gas and/or liquid introduced into the apparatus 10b. Where the apparatus 10b includes multiple rigid walls, the apparatus 10b can be inflated to different extents such that the walls form different angles relative to each other. For instance, each rigid wall can form a unique, desired angle with an adjacent non-rigid wall. FIGS. 38 and 39 also show that the apparatus 10b can in some cases be completely or substantially flattened when fully deflated, e.g., to permit compact storage of the apparatus 10b.

In some embodiments, the apparatus 10b is empty and void of any internal structure. In other embodiments, the apparatus 10b has inner chambers to provide rigidity, provide ballast and/or help define an inflated shape of the apparatus 10b. Additionally, the apparatus 10b can include internal structural elements that are independent of gas or liquid. Such internal structure could include drop stitch technology, in which tethers, such as strings or ropes, are attached to two or more walls 615 that define the shape of the apparatus 10b. These tethers can optionally be flexible in a non-pressurized compressed environment. However, when the apparatus 10b is inflated, the tethers may become taut and prevent the walls 615 from extending beyond an intentional length, thus confining the apparatus 10b to a desired pressurized shape.

As shown in FIG. 40, the apparatus 10b is used in combination with a boat 50 and can be attached to various locations on the boat 50. For example, the apparatus 10b can be attached to a single side 70, 80 of the hull 40. Alternatively, two inflatable apparatuses 10b can be used such that one apparatus 10b is attached to one side 70 of the hull 40 while another apparatus 10b is attached to the other side 80 of the hull 40. Each apparatus 10b can be permanently or removably attached to the boat 50. Preferably, the apparatus 10b is attached to the boat 50 by at least one, and preferably two, suction cups 620. This, however, is by no means required. It is also contemplated that the apparatus 10b can include only a single suction cup 620, or more than two suction cups 620. Alternatively, other removable and non-invasive forms of mechanical attachment, such as magnets,

18

hook and loop fasteners or the like, can be provided for securing the apparatus 10b to the hull 40 of a boat 50.

Where two inflatable apparatuses 10b are used on a boat 50, each apparatus 10b can have an independent source (or a common source) of gas and/or liquid as the means for inflation. As shown in FIG. 40, this source can include a controller 625 to regulate a volume of gas and/or liquid distributed into each apparatus 10b by increasing or decreasing the pressure of the gas and/or liquid. These pressure changes can occur with the use of conventional valves in either an open or contained system. Such a system could have one or more gas or liquid pumps to operate the system to achieve desired levels. The system could also be an intake and dump-type siphon system, in which movement of the boat 50 is used as a means of intake to pressurize the system. Deflating or dumping pressure can be accomplished, for example, by pump or generator, natural aspiration allowing pressure inside the source container to equalize naturally, and/or with the assistance of motion of the boat 50 and force inflicted on the outside of the source containers.

Such a system, as described above, would allow switching from operation of one inflatable apparatus 10b on one side 70 of the hull 40 to operation of the other apparatus 10b on the opposite side 80 of the hull 40 as needed. Thus, a selected one or more of the inflatable apparatuses 10b can be inflated, or all of the inflatable apparatuses 10b can remain in their uninflated states. Advantageously, this system creates the desired effect of either virtually elongating one side 70, 80 of the hull 40 or intentionally destroying wake development on one side of the boat 50. Both effects can be used to generate mismatched wake conversion, and therefore, larger wake development for one side of the boat 50 relative to the other side of the boat 50.

In certain embodiments, the inflatable apparatus 10b has at least one wall 615 that is concave. In the embodiment of FIGS. 41 and 42, for example, a wall 615 (e.g., a top or leading wall) is concave so as to bound a channel 615C. Configurations of this nature may provide special wake effects, e.g., by virtue of creating a somewhat channeled flow, which may have particular flow characteristics, such as laminar or substantially laminar flow through the channel. When the inflatable apparatus 10b is mounted operatively to the hull of a boat, the wall 615 bounding the channel 615C may face generally away from the hull and/or it may be oriented at an angle α (as measured relative to an axis normal to the hull surface to which the apparatus is attached and/or relative to a bottom wall 615 of the apparatus). The angle α convention here is best appreciated with reference to FIG. 39.

As shown in FIG. 39, the inflatable apparatus 10b can optionally include a leading wall 615 and a trailing wall 615 that are oriented at different angles α , θ . In some cases, angle α may be greater than 45 degrees, while angle θ is less than 45 degrees.

The inflatable apparatus 10b preferably has a wall 615 (e.g., a bottom wall) to which one or more (e.g., a plurality of) suction cups 620 are attached. In some cases, the one or more suction cups 620 are joined to a bottom wall 615 of the inflatable apparatus 10b by ultrasonic welding. In such cases, the suction cups are not intended to be separated from the bottom wall of the apparatus, but rather are integral/permanently attached to the bottom wall.

In some embodiments, the inflatable apparatus 10b includes a first wall 615 (e.g., a leading or front wall) that is rigid, a second wall 615 (e.g., a bottom wall) that is rigid, and one or more walls (e.g., side walls) 615 that are flexible. The one or more flexible walls may form (or be part of) a

flexible, inflatable/collapsible enclosure, such as a PVC bag or another flexible bag or other bellows. The second wall 615 may be a bottom wall, and one or more suction cups may be attached to the bottom wall. Furthermore, if an apparatus of this nature has a channel 615C, then the first wall 615 can optionally bound the channel.

In embodiments where the wake shaping apparatus 10 is an inflatable apparatus 10b, one or more of the walls 615 may be rigid. As just one example, such a rigid wall may comprise flexible top and bottom layers (e.g., comprising PVC or another polymer) between which there is sandwiched a rigid sheet (e.g., comprising ABS or another rigid plastic).

FIG. 43 shows another embodiment involving an inflatable apparatus 10b. Here, the wake shaping apparatus 10 comprises a base 10bp that can be removably secured to the hull 40 of a boat, and an inflatable apparatus 10b that can be removably secured to the base 10bp. The inflatable apparatus 10b may be removably secured to the base 10bp by virtue of, for example, the bottom wall 615 of the inflatable apparatus 10b having two slots respectively configured to releasably receive and retain the two levers 195. Additionally or alternatively, there can be another releasable mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like.

As shown in FIG. 43, the base 10bp preferably includes one or more (preferably a plurality of) suction cups 620. In the embodiment illustrated, each suction cup 620 is operably coupled with a lever 195, which can optionally be of the nature described elsewhere in this disclosure. Instead of levers, the suction cups can have other actuators, or may simply be configured to be pressed onto the hull without having actuators. Furthermore, instead of suction cups, the base can be removably attached to the hull through other non-invasive forms of mechanical attachment, such as hook and loop fasteners, clasps, magnets, or the like. Moreover, in some cases, the base 10bp can have a different configuration so as to be attached permanently to the hull.

In certain embodiments, the inflatable apparatus 10b includes a wall 615 that is a leading (or "front") wall, and thus faces generally way from the bottom of the apparatus and/or away from the hull 40. If desired, this leading (or "front") wall 615 can be a rigid wall having a flange 615F projecting rearwardly of a rear wall (which may be a flexible, collapsible/expandable rear wall) of the inflatable apparatus. Reference is made to FIG. 43. While this figure involves a separable base 10bp, it is to be appreciated that a rearwardly projecting flange 615F can optionally be provided on an embodiment like the one shown in FIGS. 41 and 42, or on various other inflatable embodiments where there is no separable base.

In some embodiments, the inflatable apparatus 10b has one or more apertures 615H formed so as to extend into the apparatus. Reference is made to FIGS. 44A-44C. Here, one or more apertures 615H pass entirely from (and through) one wall 615 (optionally a sidewall) of the inflatable apparatus 10b to (and through) another wall 615 (optionally an opposed sidewall) of the apparatus 10b. In other cases, the apertures may be formed as blind bores. The aperture(s) 615H preferably are exposed/open through one or more walls 615 of the inflatable apparatus 10b. In some cases, each aperture 615H opens through two walls 615 of the inflatable apparatus 10b. When provided, the aperture(s) 615H are configured to reduce buoyancy of the inflatable apparatus 10b. This can make it easier to push under water, such as when mounting onto the hull 40 of a boat. Although not shown in FIGS. 44A-44C, there preferably are one or

more suction cups or other mechanical attachments on the bottom of the inflatable apparatus in these embodiments.

Thus, embodiments of the wake shaping apparatus are disclosed. One skilled in the art will appreciate that the present invention can be practiced with embodiments other than those disclosed. The disclosed embodiments are presented for purposes of illustration and not limitation, and the present invention is limited only by the claims that follow.

What is claimed is:

1. A wake shaping apparatus comprising a base and a blade, the base having first and second end regions and first and second side regions, the first and second side regions extending between the first and second end regions, the base having opposed top and bottom sides, the wake shaping apparatus configured such that an angular orientation of the blade is adjustable, relative to the base, by virtue of the blade being pivotable via a hinge, the wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, the base having a maximum width, measured from the first side region of the base to the second side region of the base, and each suction cup having a diameter substantially equal to the maximum width of the base, each of the suction cups having attached thereto a post, with each such post attached to a center of a respective one of the suction cups and extending outwardly therefrom along an axis, such that the suction cups have respective posts attached thereto and extending outwardly therefrom along respective axes, wherein each suction cup is coupled to a respective lever such that each lever is operable to activate and release suction for a respective one of the suction cups, each lever having a first position and a second position, such that when each lever is in the first position each suction cup is in a locked configuration, and when each lever is in the second position each suction cup is in an unlocked configuration.

2. The wake shaping apparatus of claim 1, wherein the blade projects away from the base.

3. The wake shaping apparatus of claim 1, wherein the blade has a width greater than the maximum width of the base.

4. The wake shaping apparatus of claim 1, wherein the base and the blade are attached to each other removably.

5. The wake shaping apparatus of claim 1, wherein the blade has a mount structure that projects outwardly from a face of the blade.

6. The wake shaping apparatus of claim 1, wherein the wake shaping apparatus is configured such that the angular orientation of the blade can be adjustable by pivoting the blade about a pivot axis that is parallel to the axes along which the posts extend.

7. The wake shaping apparatus of claim 1, wherein the suction cups remain in a constant plane when the angular configuration of the blade is adjusted.

8. The wake shaping apparatus of claim 1, wherein the plurality of suction cups comprises two suction cups, such that the wake shaping apparatus comprises two levers, a first of the two levers when in the second position having a free end pointing upwardly away from the base, and a second of the two levers when in the second position having a free end pointing upwardly away from the base, and wherein the first of the two levers when in the first position has its free end pointing in one direction, whereas the second of the two levers when in the first position has its free end pointing in another direction that is opposite said one direction.

9. The wake shaping apparatus of claim 1 wherein, rather than the base and the blade being attached to each other removably, the base is permanently affixed to the blade.

21

10. A wake shaping apparatus comprising a base and a blade, the base having first and second end regions and first and second side regions, the first and second side regions extending between the first and second end regions, the base having opposed top and bottom sides, the wake shaping apparatus configured such that an angular orientation of the blade is adjustable, wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, each of the suction cups having attached thereto a post, with each such post attached to a center of a respective one of the suction cups and extending outwardly therefrom along an axis, such that the suction cups have respective posts attached thereto and extending outwardly therefrom along respective axes, wherein each suction cup is coupled to a respective lever such that each lever is operable to activate and release suction for a respective one of the suction cups, each lever having a first position and a second position, such that when each lever is in the first position each suction cup is in a locked configuration, and when each lever is in the second position each suction cup is in an unlocked configuration, wherein the base comprises a pivot connection between the first and second end regions, the pivot connection comprising a ball joint.

11. A wake shaping apparatus configured to be attached removably to a hull of a boat so as to disrupt water flow around the hull such that a wake created during operation of the boat is larger, differently positioned, and/or differently shaped than when the boat is operated without the wake shaping apparatus attached to the hull, the wake shaping apparatus comprising a blade and a base that has opposed first and second end regions and opposed top and bottom sides, the wake shaping apparatus further comprising a plurality of suction cups on the bottom side of the base, each of the suction cups being part of a suction cup assembly that further includes a post, a pin, and a lever, whereby the wake shaping apparatus includes a plurality of posts, a plurality of pins, and a plurality of levers, such that each suction cup has attached thereto a respective one of the plurality of posts, each post has a hole therein, each lever has a hole therein, and each pin is received both in the hole in a respective one of the plurality of posts and in the hole in a respective one of the plurality of levers, wherein the base of the wake

22

shaping apparatus has first and second side regions, the base has a maximum width, measured from the first side region of the base to the second side region of the base, and each suction cup has a diameter that is substantially equal to the maximum width of the base, the wake shaping apparatus further including a lanyard assembly comprising a float and a float line, the float line configured to be simultaneously attached to both the base and the float, such that the wake shaping apparatus is configured to float in a body of water, the wake shaping apparatus having a lanyard eyelet configured at the first or second side region of the base, wherein the lanyard eyelet is adjustable between a first position and a second position, such that when in the first position the lanyard eyelet is at the first side region of the base, and when in the second position the lanyard eyelet is at the second side region of the base.

12. The wake shaping apparatus of claim 11, wherein the float line is configured to be attached to an opening of the lanyard eyelet having a top open side exposed at the top side of the base and a bottom open side exposed at the bottom side of the base adjacent to an outer perimeter of one of the plurality of suction cups on the bottom side of the base.

13. The wake shaping apparatus of claim 11, wherein when the wake shaping apparatus is attached removably to a hull of a boat the opening of the lanyard eyelet to which the float line is configured to be attached is spaced apart from the hull by the plurality of suction cups such that a resulting clearance between the hull and the opening facilitates attachment of the float line to the opening of the lanyard eyelet.

14. The wake shaping apparatus of claim 11, wherein the float is a closed foam material.

15. The wake shaping apparatus of claim 11, wherein the float has a cylindrical configuration with a dome-shaped leading end.

16. The wake shaping apparatus of claim 11 wherein, rather than the base and the blade being attached to each other removably, the base is integral to the blade.

17. The wake shaping apparatus of claim 11, wherein the float line comprises a rope attached directly to both the float and the lanyard eyelet.

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